

Generosity, Legibility, and Fertility Inertia

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Abstract

Large family-policy budgets have often produced surprisingly small changes in fertility. The programs that do move births tend to share a different feature: households can tell what they will receive and for how long. I study this distinction in a dynastic fertility model in which parents see public news and policy announcements but are unsure how closely either maps into their own future. A cautious parent evaluates each family plan using the least favorable interpretation she cannot rule out. This behavior creates fertility inertia: moderately adverse news reduces welfare without changing fertility, investment per child, or consumption. It also makes the perceived value of a transfer depend on its worst-case household-level pass-through rather than on the government's expenditure. An opaque program must therefore promise more than a transparent entitlement to generate the same fertility response. The analysis suggests that family policy has two distinct margins, generosity and legibility, and that increasing the first without improving the second can make fertility less responsive to subsequent news.

Keywords: Fertility; family policy; legibility; ambiguity aversion; information quality

JEL classification: J13, J18, D81

1 Introduction

South Korea offers a stark version of a familiar policy puzzle. Since 2006, the country has spent roughly 380 trillion won—about 270 billion U.S. dollars—on measures intended to raise births. The package has included birth grants, monthly child allowances, housing support for newlyweds, parental leave, childcare expansion, and subsidies for fertility treatment. Over those same years, the total fertility rate fell from 1.13 to 0.72; in Seoul, it reached 0.55. Japan has likewise added 3.6 trillion yen a year to family policy while recording nine consecutive years of record-low fertility.¹ Whatever explains low fertility, it is not a lack of policy effort.

This aggregate record can obscure an important difference across programs. Cash bonuses at birth generally raise births, but the effects are modest and often short-lived (Milligan, 2005; González, 2013; Cohen et al., 2013). South Korea's district baby bonuses provide especially careful evidence. They increased completed fertility rather than merely changing its timing, yet the estimated elasticities imply that replacement fertility would require transfers ten to sixteen times larger—roughly a median household's annual income (Kim, 2024). Other programs perform better at comparable, and sometimes lower, fiscal cost. Formula-based parental benefits

¹The cumulative Korean spending figure, roughly 380 trillion won since 2006, is the government's tally across the Basic Plans on Low Fertility and Aged Society, as reported in early 2024 (Al Jazeera, 2024). Korean fertility rates are from Statistics Korea vital statistics (Statistics Korea, 2024). Japanese birth counts are from the Ministry of Health, Labour and Welfare vital statistics (Japan Ministry of Health, Labour and Welfare, 2025), and the annual family-policy commitment refers to the acceleration plan of the Children's Future Strategy (Children and Families Agency, Government of Japan, 2023).

and a legal right to a childcare place, for example, generate larger and more persistent responses (Raute, 2019; Bauernschuster et al., 2016).

The useful distinction is not simply cash versus services or large versus small budgets. It is whether an eligible household can work out, before having a child, what the program will deliver. I call this property *legibility*. A statutory benefit formula is relatively legible. A discretionary bonus whose eligibility, duration, or local implementation may change is not. Fertility responses appear to line up more closely with this difference than with expenditure itself. The question is why the delivery of a benefit should matter so much when its expected value is the same.

The standard dynastic fertility model has no answer. In Becker and Barro (1988) and Barro and Becker (1989), a subsidy lowers the resource cost of a child. Two programs with the same expected value then have the same effect, regardless of how households learn about them or how the benefits are administered. Inattention, adjustment costs, and slow-moving norms can all reduce the size of the response. Those frictions are plausible, and the mechanism developed here is meant to complement them. Their usual effect, however, is broad-based damping. The mechanism in this paper instead produces a particular asymmetry: parents discount favorable news, do not change births after moderately unfavorable news, react to severely bad news, and respond more to a benefit they can verify than to a larger promise they cannot.

The starting point is a practical uncertainty faced by prospective parents. A national announcement can be perfectly clear in the aggregate and still be hard to translate into one household's future. Access to childcare may depend on location and waiting lists. Leave benefits may depend on employment several years from now. Eligibility rules can change, and a policy announced by one government may not survive the next. Parents need not misunderstand the announcement to remain unsure about its relevance to them.

I place that uncertainty inside a recursive Barro–Becker model. Parents choose fertility and investment per child after observing public news about the environment their descendants will face. They understand the signal but do not know how informative it is for their own dynasty. For each candidate family plan, a cautious parent uses the least favorable informativeness parameter in a set of values she considers possible. The adverse interpretation is therefore chosen plan by plan; it is not an exogenous pessimistic belief imposed on every decision.

Why can this produce inaction? Bad news has two effects that pull in opposite directions. It lowers the expected continuation value of a child. At the same time, an informative signal resolves some of the risk shared by all children in the family. Shared risk matters because siblings do not diversify across different housing markets, labor markets, or policy regimes. A larger family is a larger exposure to one common future. When news is moderately bad, treating it as highly informative makes its content more damaging but leaves less unresolved risk. Treating it as weakly informative softens the bad news but leaves more risk in place. The least-favorable balance between those two considerations varies with the family plan.

In the closed-form model, that adjustment exactly offsets the signal's effect on the marginal value of another child over an interval of moderately adverse news. Welfare falls throughout the interval, but fertility, investment per child, and consumption do not move. I refer to this separation between welfare and choice as *fertility inertia*. It is not inattention: the parent processes the news, and her assessment of the future deteriorates. The adjustment takes place in the interpretation of the signal rather than in the number of births.

Outside the inertia interval, the response is asymmetric. Favorable news is evaluated under the possibility that it says little about the household's future. Very unfavorable news is evaluated as highly informative. The model therefore gives modest responses to good news, no behavioral response to moderately bad news, and a sharper response to sufficiently bad news. Both risk aversion and ambiguity about signal informativeness are needed for this pattern. With ambiguity alone, beliefs are pessimistic but there is no inertia interval. With risk aversion alone and a known signal process, fertility changes smoothly with the signal.

The same reasoning changes the value of family policy. Let a per-child subsidy have a known

statutory amount but an uncertain household-level pass-through. The uncertainty can reflect future eligibility, take-up, administrative discretion, or the durability of the program. If $\underline{\alpha}$ is the smallest pass-through the household cannot rule out, its fertility choice responds to $\underline{\alpha}$ times the statutory benefit. A fully transparent entitlement and an opaque promise of the same size are therefore not equivalent. The opaque program must be multiplied by $1/\underline{\alpha}$ to produce the same fertility response, even though the government pays according to the pass-through that ultimately occurs. As $\underline{\alpha}$ approaches zero, additional statutory generosity has almost no effect on fertility. The model also yields a less immediate result: increasing an uncertain benefit can widen the range of news to which fertility does not respond.

This separation between generosity and legibility changes how policy should be compared. A larger benefit raises the average payoff to parenthood. Clear eligibility rules, guaranteed childcare places, enforceable job-return rights, and credible multi-year commitments do something different: they make that payoff easier for a household to verify. Such reforms raise the value of the spending already in place and, when they also clarify how public conditions map into a household's future, can make fertility more responsive to later information.

The mechanism has observable implications. First, fertility should react more to severely bad news than to comparably sized good news, with little or no movement after moderately bad signals. Second, subjective assessments can worsen even when births do not; intentions and sentiment record the welfare effect that realized fertility may miss. Third, programs should be ranked by verifiability per dollar rather than by fiscal size alone. Finally, a reform that makes existing benefits easier to verify can affect births without raising expenditure. Section 2 reviews evidence for the first three claims. The fourth remains, as far as I know, untested.

Related literature The paper speaks first to the economics of fertility. The quantity–quality tradition begins with [Becker and Lewis \(1973\)](#); [Razin and Ben-Zion \(1975\)](#) provided an early dynamic optimizing treatment, and [Becker and Barro \(1988\)](#); [Barro and Becker \(1989\)](#) placed fertility in a dynastic framework. Later work embedded endogenous fertility in models of growth, human capital, and inequality ([Galor and Weil, 2000](#); [de la Croix and Doepke, 2003](#); [Doepke and Zilibotti, 2005](#); [Manuelli and Seshadri, 2009](#)), while another line studied income risk, employment risk, fecundity risk, and incomplete markets ([Ranjan, 1999](#); [Sommer, 2016](#); [Doepke et al., 2023](#)). These models differ in many respects, but they ordinarily give the household one known probability law. Here the object of uncertainty is the meaning of public information for a particular dynasty. That addition produces the inertia result and distinguishes a program's promised generosity from the amount a household is willing to count on.

This distinction is motivated by the small benefit elasticities found for cash transfers. [Kim \(2024\)](#) estimates elasticities of completed fertility between 0.03 and 0.06, and [Hart et al. \(2024\)](#) reach a similar conclusion in their review of policy evaluations. Demographic research also emphasizes subjective uncertainty about one's own future, rather than average conditions alone, as a constraint on fertility intentions ([Vignoli et al., 2020](#); [Guetto et al., 2022](#)). In the model, ambiguity absorbs part of a public signal before it reaches the fertility margin.

The second connection is to decision-making under ambiguity. The formal ingredients come from maxmin expected utility ([Gilboa and Schmeidler, 1989](#)), its recursive formulation ([Epstein and Schneider, 2003](#)), smooth ambiguity ([Klibanoff et al., 2005, 2009](#)), and robust control ([Hansen and Sargent, 2001, 2008](#)). The closest precedents study information quality in financial markets. [Epstein and Schneider \(2008\)](#) show that investors who are unsure about signal precision use a worst-case assessment and respond more to bad news than to good news. Related equilibrium and learning results appear in [Caskey \(2009\)](#) and [Ju and Miao \(2012\)](#). [Illeditsch et al. \(2021\)](#) derive information inertia when risk and ambiguity jointly make portfolios unresponsive to worse-than-expected news.

A fertility choice differs from a portfolio choice in ways that strengthen the role of shared risk. A child is a long-lived, nontraded claim whose payoff is an endogenous dynastic continuation

Table 1: Fertility responses to family policy, by program legibility

Program (study)	Instrument	Legibility	Fertility response
Quebec birth bonus (Milligan, 2005)	cash at birth	Low (ended 1997)	positive, then faded
Spain baby bonus (González, 2013)	cash at birth	Low (abrupt start and end)	modest for its size
Israel child subsidies (Cohen et al., 2013)	child allowance	Medium (often revised)	positive for some
South Korea baby bonus (Kim, 2024)	cash at birth	Mixed (simple claim, locally set, renewed annually)	positive, small per unit
Germany parental benefit (Raute, 2019)	earnings-linked benefit	High (fixed formula)	sizable, persistent
Germany childcare (Bauernschuster et al., 2016)	guaranteed slot	High (legal claim)	persistent per euro

Notes: Qualitative summary of the cited studies’ headline findings, in the studies’ own terms. The legibility rating is ours. An instrument is legible when an eligible household can verify, ex ante, what it will receive and for how long.

value. Children in one family also share much of the same future; adding a child concentrates rather than diversifies the exposure. Moving information-quality ambiguity into this setting turns a forgone trade into a forgone birth and connects ambiguity over public information to the design of family policy.

Outline Section 2 presents the empirical patterns that motivate the model. Section 3 sets out the recursive environment and derives the general optimality conditions. Section 4 establishes fertility inertia, introduces family policy, and considers smooth ambiguity and integer fertility. Section 5 concludes.

2 Motivating Evidence

I use three descriptive patterns to discipline the theory. They are not presented as new causal estimates. Rather, each points to a feature that a useful model should be able to reproduce.

2.1 Fact 1. Legible programs move births more per unit of fiscal cost

The first pattern comes from quasi-experimental evaluations of family policy. Table 1 organizes several well-identified studies by a feature that the original evaluations do not usually emphasize: how easy it was for an eligible household to verify the benefit in advance. I classify an instrument as legible when a household deciding whether to have a child could determine both what it would receive and how long the benefit would last.

The comparison is suggestive. The larger and more durable responses per unit of expenditure come from benefits that operate as verifiable entitlements. Responses to cash bonuses are smaller and often disappear after the program is revised or withdrawn. South Korea’s district baby bonus, the cleanest case in the table, makes the contrast concrete. It did raise completed fertility, but estimated benefit elasticities of 0.03 to 0.06 imply that replacement fertility would require transfers ten to sixteen times as large—approximately a median household’s annual income (Kim, 2024). Earlier panel estimates point in the same direction (Hong et al., 2016).

The administration of the Korean program also created uncertainty that its statutory amount alone does not capture. District governments financed the bonus, local councils renewed it each year, and 2015 payments ranged from zero to KRW 18.8 million across districts (Kim, 2024). A household could observe the current schedule without knowing what would remain available over the years relevant to raising a child. Proposition 5 formalizes this distinction. Fertility responds

to $\underline{\alpha}\tau$, the promised benefit evaluated at its worst-case pass-through, whereas public expenditure is determined by the pass-through that eventually occurs. A low value of $\underline{\alpha}$ therefore produces exactly the combination seen in the data: a costly program and a small behavioral response.

2.2 Fact 2. Births respond asymmetrically to news

The second pattern is directional. Fertility often falls sharply after bad macroeconomic news, while favorable policy announcements generate little movement relative to their scale. South Korean births dropped after the 1997–98 financial crisis and again during the 2015–2021 rise in housing costs; cross-country studies likewise link fertility to recessions and unemployment risk. Over the same period, successive Korean Basic Plans on Low Fertility—announced in 2006, 2011, 2016, and 2019 and expanded thereafter—committed substantial public funds. The local transfers that accompanied those plans had positive causal effects, but the changes were small relative to their cost (Kim, 2024). The national fertility rate nevertheless fell from 1.24 in 2015 to 0.72 in 2023. Japan’s 2023 “different dimension” childcare package was similarly followed by another record-low birth count in 2024. These aggregate comparisons are not causal estimates; they illustrate the asymmetry that also appears in the program evaluations summarized above.

Corollary 2 gives this pattern a precise form. The parent evaluates good news using ρ , so choices move only slightly. Sufficiently bad news is evaluated using $\bar{\rho}$ and has a larger effect. Between those cases lies the inertia interval, where the fertility response is zero. A symmetric adjustment cost would dampen good and bad news in the same way and would not generate this ordering.

2.3 Fact 3. Households report uncertainty about whether policy applies to them

The third pattern comes from what prospective parents report. When the future is difficult to quantify, people describe their choices through narratives and personal scenarios rather than explicit probabilities. Uncertainty about their own circumstances, not simply unfavorable average conditions, is associated with weaker fertility intentions (Vignoli et al., 2020). In experimental evidence, greater uncertainty about future personal economic conditions lowers intended fertility even when mean prospects are held constant (Guetto et al., 2022). Among young Korean adults, intentions are also strongly related to trust in social and political institutions after accounting for economic circumstances (Kwag et al., 2024). In a system of locally determined and annually renewed benefits, such distrust naturally includes doubt about whether a policy will persist and whether a particular household will remain eligible (Kim, 2024).

These findings correspond to the two ambiguity sets in the model: $[\underline{\rho}, \bar{\rho}]$ describes uncertainty about what public news means for the household, and $[\underline{\alpha}, \bar{\alpha}]$ describes uncertainty about the delivery of an announced benefit. They also speak to a less obvious model prediction. Within the inertia interval, welfare declines although births do not. Intentions and sentiment can therefore react to information that does not yet appear in realized fertility.

Taken together, the evidence presents a fairly specific target. Verifiable programs appear to move births more effectively; fertility reacts asymmetrically to public information; and households locate much of their uncertainty in their own exposure to policy and economic conditions. The model that follows asks whether one mechanism can account for all three.

3 A Recursive Barro–Becker Model with Ambiguous Signals

I now put the argument into a recursive Barro–Becker environment. A parent observes news about the conditions her children are likely to face, but she is unsure how closely that news applies to her own dynasty. Two results will matter later. First, a range of moderately unfavorable signals changes the parent’s welfare without changing fertility, consumption, or investment per

child. Second, uncertainty about the delivery of a subsidy makes its household value depend on a worst-case pass-through rather than on the headline amount.

The source of inertia can be seen before writing down the recursion. Informative bad news lowers the expected payoff from having children, but it also resolves uncertainty about a future that siblings share. A parent who is unsure about informativeness chooses the least favorable reading for each family plan. For moderately bad news, the loss in the mean and the gain from risk resolution offset at the margin. Because the adverse reading is allowed to vary with the plan, the offset is not confined to one candidate choice.

Formally, the model combines a Barro–Becker dynasty, a CARA certainty equivalent for risky continuation values, and maxmin ambiguity over signal informativeness (Hansen and Sargent, 2001; Gilboa and Schmeidler, 1989). I first show that the recursion is well defined and derive its optimality conditions (Propositions 1–2). The local normal representation in Section 4 then yields the inertia interval and a closed-form expression for its width.

3.1 Environment

The state. Time is discrete and each period corresponds to one generation. A parent enters the period with state

$$(x_t, s_t) \in \mathcal{X} \times \mathcal{S}.$$

The vector x_t contains the payoff-relevant household and aggregate state: assets, human capital, age, economic conditions, and policy variables. The scalar s_t is public news about the environment in which the next generation would grow up. It may concern housing costs, childcare, labor markets, or family policy. Larger values of s_t represent better news. Current resources are $m_t \equiv y_t + a_t$, with income y_t and assets a_t determined by x_t .

The controls. The controls are consumption $c_t \geq 0$, fertility $n_t \geq 0$, and a transfer or investment $z_t \geq 0$ for each child. The last control is the quality margin in the familiar quantity–quality tradeoff. I treat fertility as continuous; a fractional value can be interpreted as the intensity or timing of an additional-birth plan. The budget constraint is

$$c_t + n_t(q_t + z_t) \leq m_t, \tag{1}$$

where $q_t > 0$ is the direct cost of a child. Total resources devoted to each child are therefore $q_t + z_t$.

The transition and the ambiguous parameter. The next generation’s state is generated by

$$(x_{t+1}, s_{t+1}) = G(x_t, s_t, z_t, \varepsilon_{t+1}; \rho), \tag{2}$$

The innovation ε_{t+1} is realized after the parent chooses and has a distribution independent of those choices. The parameter ρ indexes the probability models the parent regards as possible. In particular, it measures how strongly the public signal predicts the next generation’s continuation value. The parent knows that the signal is relevant but cannot pin down its informativeness:

$$\rho \in [\underline{\rho}, \bar{\rho}], \quad 0 < \underline{\rho} < \bar{\rho} < 1, \tag{3}$$

and assigns no prior over this interval. Aggregate news about housing or childcare is plainly relevant to parents as a group, yet a household may not know how well it predicts the circumstances of its own descendants. The uncertainty over ρ is Knightian; it is separate from ordinary risk over ε_{t+1} . I write expectations under model ρ and investment z_t as $\mathbb{E}_{\rho, z_t}[\cdot \mid x_t, s_t]$, suppressing subscripts when they are clear.

Timing. The timing is straightforward. The parent first sees s_t , then chooses (c_t, n_t, z_t) , and finally observes ε_{t+1} . Thus the choice is made with risk over the innovation and ambiguity over its signal-loading parameter ρ .

The dynastic objective. Let

$$V' \equiv V(x_{t+1}, s_{t+1})$$

denote a child's lifetime continuation value, including her own future fertility choices. The same value function applies in every generation, so the return to a child is endogenous rather than a primitive payoff. Fertility scales that continuation value through the Barro–Becker factor $\Phi(n_t)$, where

$$\Phi(0) = 0, \quad \Phi'(n) > 0, \quad \Phi''(n) \leq 0.$$

With no children, the dynasty receives no continuation payoff. Additional children increase dynastic weight, though weakly at a decreasing rate. A standard specification is $\Phi(n) = n^{1-\eta}/(1-\eta)$ for $\eta \in [0, 1)$; the case $\eta = 0$ gives $\Phi(n) = n$. The results do not rely on this particular form.

3.2 Risk-sensitive continuation value

When the parent chooses, V' is still random. I summarize that risk with an exponential certainty equivalent. For any continuation payoff Y , define

$$\mathcal{C}(Y \mid x_t, s_t) = \mathcal{C}_{\gamma, \rho, z_t}(Y \mid x_t, s_t) \equiv -\frac{1}{\gamma} \log \mathbb{E}[\exp(-\gamma Y) \mid x_t, s_t], \quad \gamma > 0. \quad (4)$$

This is the sure payoff that a CARA decision maker with absolute risk aversion γ considers equivalent to Y . Greater γ places a larger penalty on risk, while the limit as $\gamma \rightarrow 0$ is the ordinary expectation. The operator is standard in risk-sensitive control ([Hansen and Sargent, 1995](#); [Tallarini, 2000](#)). For normal payoffs it has the exact mean–variance form stated next.

Lemma 1 (CARA certainty equivalent). *$\mathcal{C}_{\gamma, \rho, z_t}(Y \mid x_t, s_t)$ is the sure payoff a CARA agent with utility $U(Y) = -\exp(-\gamma Y)$ regards as equivalent to Y under model ρ . If $Y \mid x_t, s_t, \rho \sim N(\mu_Y, \sigma_Y^2)$, the formula is exact and linear–quadratic,*

$$\mathcal{C}(Y \mid x_t, s_t) = \mu_Y - \frac{\gamma}{2} \sigma_Y^2.$$

For general Y whose cumulant-generating function $\kappa(t) = \log \mathbb{E}e^{tY}$ is finite and twice differentiable in a neighborhood of $t = 0$, a small- γ expansion gives $\mathcal{C}(Y) = \mathbb{E}[Y] - \frac{\gamma}{2} \text{Var}(Y) + o(\gamma)$, so $\lim_{\gamma \rightarrow 0} \mathcal{C}(Y) = \mathbb{E}[Y]$.

Appendix [A.1](#) gives the derivation. Under normality, risk subtracts $\gamma/2$ times the variance from the mean. The same expression is the leading approximation for a general payoff when γ is small.

The recursive problem. The payoff to fertility is the scaled continuation value

$$Y_{t+1}(n_t, z_t) = \Phi(n_t)V(x_{t+1}, s_{t+1}). \quad (5)$$

The parent evaluates this payoff in two stages. She first computes its certainty equivalent under a particular model and then takes the least favorable value over the admissible models. This is the maxmin ordering of [Gilboa and Schmeidler \(1989\)](#): each action is judged under its own worst plausible probability law. The recursive problem is

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathcal{C}_{\gamma, \rho, z_t}[\Phi(n_t)V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\} \quad (6)$$

subject to (1). Flow utility satisfies $u' > 0$ and $u'' < 0$, and $\beta \in (0, 1)$. The Barro–Becker fixed point remains intact: the value of a child is still the next generation’s value function. Only the evaluation of that value changes. The order of the two operators is important. Nature selects ρ separately for each proposed (n_t, z_t) , allowing the least-favorable reading of the news to change with the family plan.

Interpretation The certainty equivalent captures unavoidable risk. The minimization over ρ captures uncertainty about how to read the signal. These two sources of caution interact differently with different plans. A large family has more exposure to the common future and therefore more to lose from unresolved risk; its adverse reading of moderately bad news places less weight on the signal. For a smaller family, the mean effect matters more, so the adverse reading is more informative. If the pessimistic belief were fixed, the two effects would balance at only one plan. Here the belief moves with the plan, allowing the balance to persist over an interval and creating a flat policy region.

3.3 Recursive problem and solution concept

The value function appears on both sides of (6), so existence and uniqueness are not automatic. Define the Bellman operator T by

$$(TV)(x, s) = \max_{c, n, z} \left\{ u(c) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathcal{C}_{\gamma, \rho, z} [\Phi(n)V(x', s') \mid x, s] \right\} \quad (7)$$

subject to

$$c + n(q(x, s) + z) \leq m(x, s).$$

A solution is a fixed point $V = TV$ together with policy functions

$$c^*(x, s), \quad n^*(x, s), \quad z^*(x, s),$$

and a worst-case belief policy

$$\rho^*(x, s) \in [\underline{\rho}, \bar{\rho}].$$

Assumption 1 (Regularity). (i) $\mathcal{X} \times \mathcal{S}$ is a Polish space. (ii) Per-child investment is bounded, $z \in [0, \bar{z}]$, so that the feasible correspondence $\Gamma(x, s) = \{(c, n, z) \geq 0 : c + n(q(x, s) + z) \leq m(x, s), z \leq \bar{z}\}$ is nonempty, compact-valued, and continuous. (iii) u is continuous, increasing, concave, and bounded on the graph of Γ , with $\sup_{(x, s), a \in \Gamma(x, s)} |u(c)| < \infty$. (iv) Φ is continuous and increasing with $\bar{\Phi} \equiv \sup_{n \in \Gamma_n} \Phi(n) < \infty$, where Γ_n is the (compact) set of feasible fertility levels, and $\beta \bar{\Phi} < 1$. (v) The transition law $P_{\rho, z}(\cdot \mid x, s)$ is weakly continuous (Feller) in (x, s, z) and continuous in ρ . The law of the innovation ε_{t+1} does not depend on the controls, so the subscript z in $\mathbb{E}_{\rho, z}$ records only that the successor state depends on z through G . The operator T acts on the Banach space of bounded continuous functions on $\mathcal{X} \times \mathcal{S}$ equipped with the sup norm.

Assumption 1 follows the recursive ambiguity framework of Epstein and Schneider (2003). One issue requires care. Repeated period-by-period minimization need not be dynamically consistent unless the set of one-step beliefs is rectangular and updated prior by prior. Treating ρ as one unknown parameter fixed forever would generally fail that requirement. I therefore state the intergenerational ambiguity set explicitly.

Assumption 2 (Generation-by-generation ambiguity). For each state and action, nature’s set of one-step-ahead transition laws is the kernel correspondence

$$\mathcal{P}(x, s, z) = \{P_{\rho, z}(\cdot \mid x, s) : \rho \in [\underline{\rho}, \bar{\rho}]\}.$$

The parent’s set of priors over future paths is the rectangular hull of this correspondence, the set of all path measures generated by measurable, possibly history-dependent selections $\rho_t(h_t) \in [\underline{\rho}, \bar{\rho}]$ from $\mathcal{P}$ at every date and history.

This rectangular structure has a direct economic interpretation. Each generation faces a new housing market, labor market, and policy regime, so the link between that generation's headlines and its private outcomes can change over time. Rectangularity makes the maxmin recursion dynamically consistent and ensures that plan-dependent worst-case beliefs come from the decision problem itself. The CARA layer is not covered by the original Epstein–Schneider theorem; I take the composed recursion (6) as the primitive preference specification, following the spirit of recursive robust control (Hansen and Sargent, 2008). Appendix A.7 discusses the issue and gives the entropy dual.

Remark 1 (No learning about informativeness). Under Assumption 2 the dynasty never learns about ρ . Nature may select a new ρ_t at every date and history, so past signals carry no information about the informativeness of future ones. This is not an incidental omission. Epstein and Schneider (2007) show that when ambiguity attaches to the quality of signals, even long histories need not shrink the set of beliefs. The smooth-ambiguity variant sketched in Section 4.7 is the natural framework for learning extensions, since it accommodates nondegenerate posteriors over ρ .

The condition $\beta\bar{\Phi} < 1$ plays the role ordinarily assigned to $\beta < 1$. Here the effective one-generation discount factor is $\beta\Phi(n)$ because a parent with more children places greater total weight on descendants. Uniformly bounding this product below one keeps the dynastic value finite and supplies the contraction.

Proposition 1 (Existence and characterization). *Under Assumption 1, T is a contraction on the bounded continuous functions with modulus at most $\beta\bar{\Phi}$, so a unique $V = TV$ exists. It is attained by policy functions solving the outer maximum and a worst-case belief $\rho^*(x, s)$ solving the inner minimum.*

Proof. The proof is available in Appendix A.2. □

The additional preference layers do not alter the contraction modulus. Both are nonexpansive in the sup norm, so T inherits the standard Barro–Becker bound $\beta\bar{\Phi}$. Ambiguity and risk sensitivity change the location of the optimum, not the existence of the fixed point.

3.4 Equilibrium characterization

The general first-order conditions retain the Barro–Becker structure. Their distinctive feature is the probability measure used to value future outcomes: it is distorted toward unfavorable states by the parent's own caution.

For a given state (x_t, s_t) , define the model- ρ certainty equivalent of a plan,

$$K_\rho(n, z; x_t, s_t) \equiv \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V(x_{t+1}, s_{t+1}) \mid x_t, s_t], \quad (8)$$

and its worst case across models,

$$J(n, z; x_t, s_t) \equiv \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n, z; x_t, s_t). \quad (9)$$

Since $u'(c) > 0$, the budget constraint binds at an interior solution, $c_t^* = m_t - n_t^*(q_t + z_t^*)$. Hence the reduced problem is

$$\max_{n, z} \{u(m_t - n(q_t + z)) + \beta J(n, z; x_t, s_t)\}. \quad (10)$$

Let

$$\rho_t^* \in \arg \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t)$$

be the worst-case signal-informativeness parameter at the optimum. The associated measure Q_t^* begins with the transition law $P_{\rho_t^*, z_t^*}$ and reweights successor states in proportion to $\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\}$. Low-value states receive more weight when $\gamma\Phi(n_t^*) > 0$. Caution therefore enters twice: ρ_t^* selects the least favorable informativeness, and the CARA operator tilts that law toward poor continuation outcomes. When $\gamma \rightarrow 0$, the tilt disappears and Q_t^* returns to $P_{\rho_t^*, z_t^*}$. Appendix A.8 records the Radon–Nikodym derivative and the equivalent entropy representation.

Assume that the transition admits the differentiable shock representation (2), with V differentiable along the relevant directions, and define the marginal effect of child investment on the child’s continuation value by

$$D_z V_{t+1} \equiv \nabla_x V(x_{t+1}, s_{t+1}) \cdot G_z^x(x_t, s_t, z_t, \varepsilon_{t+1}; \rho_t^*) + V_s(x_{t+1}, s_{t+1}) G_z^s(x_t, s_t, z_t, \varepsilon_{t+1}; \rho_t^*), \quad (11)$$

where G^x and G^s are the corresponding components of the transition. Thus $D_z V_{t+1}$ is the state-contingent marginal return to one more unit of investment in each child.

Proposition 2 (Interior first-order conditions). *Assume the transition admits the differentiable shock representation (2) and that V is differentiable. Let $D_z V_{t+1}$ be defined by (11). Suppose (c_t^*, n_t^*, z_t^*) is an interior optimum, the inner minimizer ρ_t^* is unique, and differentiation under the expectation is valid. Then*

$$\rho_t^* \in \arg \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t),$$

and, letting Q_t^* denote the twisted measure associated with (ρ_t^*, n_t^*, z_t^*) , the optimal policies satisfy

$$c_t^* = m_t - n_t^*(q_t + z_t^*), \quad (12)$$

$$u'(c_t^*)(q_t + z_t^*) = \beta \Phi'(n_t^*) \mathbb{E}_{Q_t^*} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t], \quad (13)$$

$$n_t^* u'(c_t^*) = \beta \Phi(n_t^*) \mathbb{E}_{Q_t^*} [D_z V_{t+1} \mid x_t, s_t]. \quad (14)$$

If $K_\rho(n_t^*, z_t^*; x_t, s_t)$ is differentiable in ρ , the worst-case belief also satisfies the one-dimensional optimality condition

$$\begin{cases} \partial_\rho K_\rho(n_t^*, z_t^*; x_t, s_t)|_{\rho=\rho_t^*} = 0, & \rho_t^* \in (\underline{\rho}, \bar{\rho}), \\ \partial_\rho^+ K_\rho(n_t^*, z_t^*; x_t, s_t)|_{\rho=\underline{\rho}} \geq 0, & \rho_t^* = \underline{\rho}, \\ \partial_\rho^- K_\rho(n_t^*, z_t^*; x_t, s_t)|_{\rho=\bar{\rho}} \leq 0, & \rho_t^* = \bar{\rho}. \end{cases} \quad (15)$$

Proof. The proof is available in Appendix A.3. □

The equations have the usual Barro–Becker interpretation, but marginal benefits are computed under Q_t^* . Because the inner minimizer is unique, the envelope theorem removes any derivative of the worst-case belief with respect to the plan. This is the familiar maxmin structure; an analogous business-cycle application appears in Ilut and Schneider (2014). In (13), the consumption cost of another child equals that child’s discounted dynastic value under the distorted measure, weighted by $\Phi'(n_t^*)$. In (14), an additional unit invested in every child costs n_t^* units today and returns $D_z V_{t+1}$ for each child, weighted by $\Phi(n_t^*)$.

Dividing (13) by (14) eliminates the marginal utility of consumption and gives the quantity–quality condition

$$\frac{q_t + z_t^*}{n_t^*} = \frac{\Phi'(n_t^*)}{\Phi(n_t^*)} \frac{\mathbb{E}_{Q_t^*} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t]}{\mathbb{E}_{Q_t^*} [D_z V_{t+1} \mid x_t, s_t]}. \quad (16)$$

The left side compares the resource cost of another child with the cost of raising investment per child. The right side combines the marginal-to-total dynastic weight with the distorted expected value of a child relative to the distorted return on child investment. It is the quantity–quality condition under ambiguous signal informativeness.

Equation (15) determines the adverse reading of the news. At an interior solution, a marginal change in informativeness leaves the minimized objective unchanged. At the endpoints, the worst case is the least or most informative admissible model. The inertia result arises when the interior condition continues to hold over a range of signals.

The distortion remains present in the ratio (16). Unless continuation value and the return to investment move together across states, pessimism reweights them differently. Ambiguity can therefore change the quantity–quality mix rather than merely shifting fertility up or down.

Remark 2 (Special cases). The model nests the standard benchmarks. As $\gamma \rightarrow 0$ the certainty equivalent converges to a conditional expectation (Lemma 1) and the problem becomes a risk-neutral robust Barro–Becker problem, in which $\Phi(n_t)$ factors out of the inner minimization. If instead the ambiguity set is a singleton, $\underline{\rho} = \bar{\rho} = \rho$, the problem is ambiguity-neutral but risk sensitive. Switching off both features recovers the standard Barro–Becker recursion. The explicit Bellman equations for each case are collected in Appendix A.9.

4 Signal Informativeness and Fertility Inertia

Closed-form policies are not available in the full recursion. To make the mechanism visible, I work with a local normal representation of the next generation’s value. The reduction keeps the dynastic interpretation while isolating the effect of signal informativeness. It produces the paper’s central result: welfare responds to moderately bad news even though fertility, investment, and consumption remain fixed.

4.1 Local normal representation

Around the current state, suppose

$$V(x_{t+1}, s_{t+1}) = \mu(z_t) + \sigma(z_t) \left[\rho s_t + \sqrt{1 - \rho^2} \varepsilon_{t+1} \right], \quad \varepsilon_{t+1} \sim N(0, 1), \quad (17)$$

with $\sigma(z_t) > 0$.

Equation (17) is a local representation of the endogenous continuation value, not a primitive child payoff. The functions $\mu(z_t)$ and $\sigma(z_t)$ summarize how parental investment affects its mean and its exposure to the common environment.

Two qualifications are important. Proposition 1 works on a space of bounded value functions, whereas a normal random variable is unbounded. The local model is therefore motivated by the recursion but is not derived from its existence theorem. Establishing that link would require either a larger function space with finite CARA values or a separate approximation result; I do not attempt either here. In addition, s_t changes beliefs about the future and no current object. I hold resources and child costs fixed, $\partial m / \partial s = \partial q / \partial s = 0$, and do not allow s_t to enter μ or σ directly. If bad news also reduced today’s resources, fertility could move through the budget even when the belief channel is inactive.

The parameter ρ affects both the mean and residual risk. Conditional on the signal, $\mathbb{E}[V(x_{t+1}, s_{t+1}) \mid s_t, \rho] = \mu(z_t) + \sigma(z_t)\rho s_t$ while $\text{Var}(V(x_{t+1}, s_{t+1}) \mid s_t, \rho) = \sigma(z_t)^2(1 - \rho^2)$. A so a larger ρ gives the news more influence on the conditional mean and leaves less uncertainty unresolved.

Now define the dynastic continuation payoff $Y_{t+1} = \Phi(n_t)V(x_{t+1}, s_{t+1})$. Under (17), Y_{t+1} is conditionally normal. The CARA certainty equivalent is therefore

$$K(n_t, z_t; \rho, s_t) = \Phi(n_t) [\mu(z_t) + \sigma(z_t)\rho s_t] - \frac{\gamma}{2} \Phi(n_t)^2 \sigma(z_t)^2 (1 - \rho^2). \quad (18)$$

The first term is expected dynastic value and the second is its risk penalty. The penalty contains $\Phi(n_t)^2$ because children share the shock. Adding children increases exposure to one common future rather than diversifying across independent outcomes.

Using (18), the parent's reduced problem is

$$\max_{n,z} \left\{ u(m_t - n(q_t + z)) + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K(n, z; \rho, s_t) \right\}. \quad (19)$$

The reduced problem retains the ordering that matters: the parent chooses fertility and investment, evaluates continuation value at the least favorable informativeness, and pays a penalty for unresolved common risk. Fixing $z = \bar{z}$ gives the two-period specialization used for the closed form in Section 4.5; its child payoff should still be read as a stand-in for endogenous continuation value.

4.2 Benchmark with known signal informativeness

Begin with known ρ . Holding the realization s_t fixed, greater informativeness need not move fertility in one direction. It amplifies the content of bad news but also removes residual risk. Proposition 3 separates those effects.

Proposition 3 (Known signal informativeness). *Fix $z_t = z$ and suppose the fertility problem has a unique interior solution $n^*(\rho, s_t)$ satisfying the second-order condition. Assume $\Phi'(n^*) > 0$ and $\sigma(z) > 0$. Under the local normal representation (17),*

$$\text{sgn} \left(\frac{\partial n^*}{\partial \rho} \right) = \text{sgn} [s_t + 2\gamma\Phi(n^*)\sigma(z)\rho]. \quad (20)$$

Thus higher signal informativeness raises fertility if and only if

$$s_t > -2\gamma\Phi(n^*)\sigma(z)\rho.$$

With endogenous z_t , let H be the Hessian of the objective with respect to (n, z) , evaluated at the optimum, and suppose H is negative definite. Then

$$\begin{pmatrix} \frac{\partial n^*}{\partial \rho} \\ \frac{\partial z^*}{\partial \rho} \end{pmatrix} = -H^{-1} \begin{pmatrix} \beta\Phi'(n^*)\sigma(z^*)[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho] \\ \beta\Phi(n^*)\sigma_z(z^*)[s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho] \end{pmatrix}. \quad (21)$$

Proof. The proof is in Appendix A.4. □

For good news, the mean and risk-resolution effects point in the same direction, so fertility rises with ρ . For bad news they conflict. Risk resolution dominates when $-2\gamma\Phi(n^*)\sigma(z)\rho < s_t < 0$; the adverse mean effect dominates below that range.

When investment is endogenous, the same scalar factor governs both policy responses in (21). At zero, neither fertility nor investment responds locally to a change in known informativeness. Under ambiguity, nature selects ρ by balancing these same forces against the parent.

Figure 1 compares known and ambiguous informativeness. With known ρ , fertility changes smoothly with the signal. Under ambiguity, very bad news is evaluated at $\bar{\rho}$, good news at $\underline{\rho}$, and fertility is flat between them. The right panel displays the adjustment in the adverse belief. As news becomes less bad, ρ_t^* falls: the parent exchanges the possibility that bad news is highly relevant for the possibility that more risk remains unresolved. Fertility does not move until the belief reaches an endpoint.

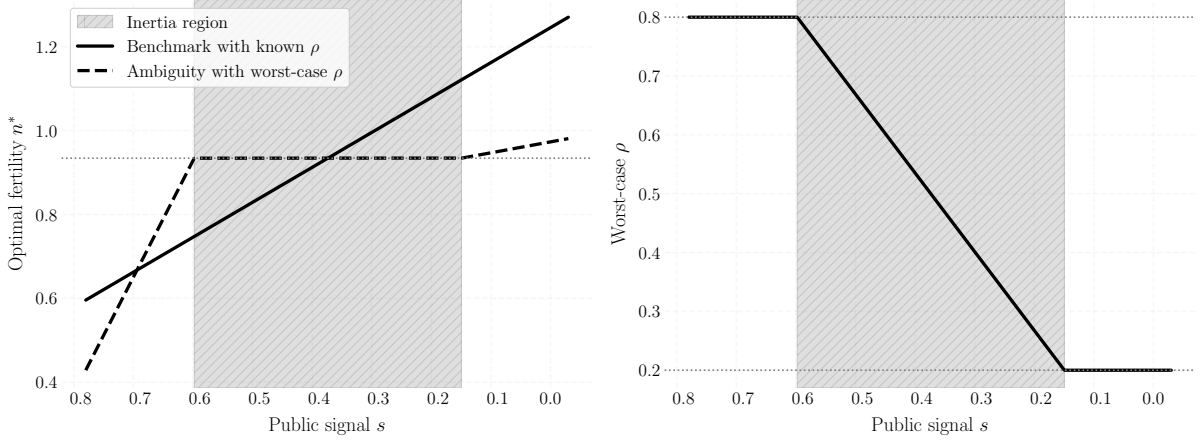


Figure 1: Optimal fertility under known and ambiguous signal informativeness

Notes: The figure compares optimal fertility when signal informativeness is known with optimal fertility when the parent evaluates the signal under the worst admissible informativeness. Parameters are $u(c) = c$, $\Phi(n) = n$, $y = 10$, $q = 1$, $\beta = 0.95$, $\mu = 1.8$, $\sigma = 1$, $\gamma = 0.8$, $\underline{\rho} = 0.2$, and $\bar{\rho} = 0.8$. The known- ρ benchmark uses $\rho = 0.5$. Under these values, the inertia fertility level is $n^0 = 0.934$, and the inertia region is approximately $s \in [-0.598, -0.149]$.

4.3 Ambiguous signal informativeness

I now allow the parent to be unsure how informative the public signal is. Fix a plan (n, z) with $\Phi(n) > 0$. (If $n = 0$, continuation value is zero and K no longer depends on ρ , so the formulas that follow are irrelevant.) For every other plan, the parent uses the least favorable admissible value of ρ and therefore evaluates $\min_{\rho \in [\underline{\rho}, \bar{\rho}]} K(n, z; \rho, s_t)$. Using (18), the derivative with respect to ρ is

$$K_\rho(n, z; \rho, s_t) = \Phi(n)\sigma(z)s_t + \gamma\Phi(n)^2\sigma(z)^2\rho,$$

and

$$K_{\rho\rho}(n, z; \rho, s_t) = \gamma\Phi(n)^2\sigma(z)^2 > 0.$$

Thus K is strictly convex in ρ . Its minimizer is the unconstrained solution, truncated when necessary to remain inside the ambiguity set:

$$\rho^w(n, z, s_t) = \text{Proj}_{[\underline{\rho}, \bar{\rho}]} \left(-\frac{s_t}{\gamma\Phi(n)\sigma(z)} \right). \quad (22)$$

Suppose $0 \leq \underline{\rho} < \bar{\rho}$. For good news ($s_t \geq 0$), the unconstrained solution is nonpositive, so the parent settles on $\rho^w = \underline{\rho}$ and gives the signal as little weight as the model permits. For sufficiently bad news, $s_t \leq -\gamma\Phi(n)\sigma(z)\bar{\rho}$, the opposite boundary binds: $\rho^w = \bar{\rho}$. The parent then treats the news as highly informative.

Between these two cases, for

$$-\gamma\Phi(n)\sigma(z)\bar{\rho} < s_t < -\gamma\Phi(n)\sigma(z)\underline{\rho},$$

the worst case is interior so that

$$\rho^w(n, z, s_t) = -\frac{s_t}{\gamma\Phi(n)\sigma(z)}.$$

This middle range is where the mechanism does its work. Here the least-favorable interpretation varies with both the news and the plan being considered. Planning a larger family, or choosing an investment that raises $\sigma(z)$, changes which value of ρ is most damaging. The result is asymmetric: the parent discounts favorable news and gives adverse news more credence. Fertility inertia arises in the interior range, where that pessimistic interpretation can move freely.

Proposition 4 (Fertility inertia). *Let (n^0, z^0) be the unique interior maximizer of the signal-free problem*

$$\max_{n,z} \left\{ u(m_t - n(q_t + z)) + \beta \left[\Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 \right] \right\}. \quad (23)$$

Suppose the objective in (23) is strictly concave in a neighborhood of (n^0, z^0) , with $\Phi(n^0) > 0$ and $\sigma(z^0) > 0$. If the signal satisfies

$$-\gamma\Phi(n^0)\sigma(z^0)\bar{\rho} < s_t < -\gamma\Phi(n^0)\sigma(z^0)\underline{\rho}, \quad (24)$$

then the worst-case signal informativeness is interior at (n^0, z^0) , and the optimal policies in (19) satisfy, locally,

$$n^*(s_t) = n^0, \quad z^*(s_t) = z^0, \quad c^*(s_t) = m_t - n^0(q_t + z^0). \quad (25)$$

Thus, throughout the inertia region, optimal fertility, child investment, and consumption are locally independent of the public signal.

Proof. The proof is available in Appendix A.5. □

The proposition is deliberately local. It identifies (n^0, z^0) as the only nearby optimum among plans with an interior worst-case belief. The same pair is globally optimal if every candidate plan has an interior worst-case belief and (23) is strictly concave over the full choice set; these conditions exclude rival optima at $n = 0$ or at a boundary value of ρ . The endpoints in (24) require a little more care. At either endpoint, the worst-case belief just touches the edge of the ambiguity set. A nearby deviation then receives a small convex gain, so the strengthened second-order condition in the appendix is needed to ensure that local concavity still dominates. In the one-dimensional example in Section 4.5, this condition is automatic.

4.4 The welfare cost of ambiguous news

The source of the flat policy response becomes transparent once the interior minimizer in (22) is substituted into (18). The resulting continuation value is

$$\min_{\rho \in [\underline{\rho}, \bar{\rho}]} K(n, z; \rho, s_t) = \Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma}. \quad (26)$$

The first two terms are what the parent would obtain from an uninformative signal: expected dynastic value less the risk penalty associated with the unresolved common component. The last term is the welfare loss from ambiguous news. Crucially, it does not vary across plans as long as their worst-case beliefs remain interior. The news makes the parent worse off, but it does not alter the marginal return to another child or to child investment. The first-order conditions inside the inertia region are consequently identical to those of the signal-free problem. Household choices appear unresponsive even though the signal matters for welfare.

Corollary 1 (Width of the inertia region). *For a fixed signal-free optimum (n^0, z^0) , the length of the fertility-inertia interval is*

$$\gamma\Phi(n^0)\sigma(z^0)(\bar{\rho} - \underline{\rho}). \quad (27)$$

Thus the direct width of the inertia region is larger when dynastic risk sensitivity γ , fertility exposure $\Phi(n^0)$, continuation-value exposure $\sigma(z^0)$, or ambiguity about signal informativeness $\bar{\rho} - \underline{\rho}$ is larger.

Each factor in (27) has a simple role. Greater risk sensitivity makes unresolved risk more costly; higher $\Phi(n^0)$ or $\sigma(z^0)$ puts more dynastic value at stake; and a wider ambiguity set allows the pessimistic belief to adjust over a larger range. Holding the signal-free plan fixed, all three enlarge the interval on which the parent keeps that plan. If a parameter change also shifts n^0 or z^0 , those indirect effects must of course be included in the total comparative static. The flat portion lies on the bad-news side: good news is read using $\underline{\rho}$, the worst news using $\bar{\rho}$, and choices are locally constant between those cases.

4.5 Closed-form special case

To see the policy rule without the surrounding notation, consider the simplest two-period version of the model. Set $u(c) = c$ and $\Phi(n) = n$, and hold child investment fixed at $z = 0$. Then

$$\mu \equiv \mu(0), \quad \sigma \equiv \sigma(0), \quad q_t = q, \quad y \equiv m_t.$$

Here q denotes the resource cost of a child and y the parent's current resources.

In the absence of a signal, fertility solves

$$\max_{0 \leq n \leq y/q} \left\{ y - qn + \beta \left[n\mu - \frac{\gamma}{2} n^2 \sigma^2 \right] \right\}.$$

The objective is strictly concave because the quadratic term in n is negative

$$\frac{\partial^2}{\partial n^2} \left\{ y - qn + \beta \left[n\mu - \frac{\gamma}{2} n^2 \sigma^2 \right] \right\} = -\beta\gamma\sigma^2 < 0.$$

The interior first-order condition therefore gives

$$n^0 = \frac{\mu - q/\beta}{\gamma\sigma^2}. \quad (28)$$

This solution is interior when

$$\mu > q/\beta \quad \text{and} \quad qn^0 < y.$$

The first inequality requires the mean dynastic payoff from an additional child to exceed its discount-adjusted cost. The second simply ensures that the implied family size is affordable.

Define

$$\lambda \equiv \frac{\mu - q/\beta}{\sigma}.$$

I use λ as a convenient risk-adjusted measure of the attractiveness of parenthood: it is the excess mean payoff per unit of common risk. By (28),

$$\gamma\sigma n^0 = \lambda.$$

Therefore the inertia interval becomes

$$s_t \in [-\lambda\bar{\rho}, -\lambda\underline{\rho}], \quad \text{with width} \quad \lambda(\bar{\rho} - \underline{\rho}). \quad (29)$$

Equation (29) separates two sources of inertia. The first, λ , captures the risk-adjusted appeal of parenthood and hence the size of the parent's exposure. The second, $\bar{\rho} - \underline{\rho}$, measures how much latitude there is to reinterpret the signal. The flat segment disappears when ρ is known and grows linearly as the ambiguity interval widens around a fixed midpoint. Thus the relevant friction is uncertainty about informativeness, rather than merely a signal that happens to be weak.

The linear specification also yields closed forms outside the inertia region, provided that $0 < n^*(s_t) < y/q$. Extremely bad signals eventually push fertility to zero, whereas sufficiently good signals make the resource constraint bind. At either corner, fertility is flat for the mechanical reason that a constraint, rather than the ambiguity mechanism, determines the choice. When the worst-case belief is fixed at a boundary $\hat{\rho} \in \{\underline{\rho}, \bar{\rho}\}$, the interior solution is

$$n^*(s_t; \hat{\rho}) = \frac{\mu + \sigma \hat{\rho} s_t - q/\beta}{\gamma \sigma^2 (1 - \hat{\rho}^2)}.$$

Corollary 2 (Closed-form policy function). *Assume the specification of this subsection with $0 \leq \underline{\rho} < \bar{\rho} < 1$, and suppose the solution remains interior, $0 < n^*(s_t) < y/q$. Then the ambiguous-signal policy is*

$$n^*(s_t) = \begin{cases} \frac{\mu + \sigma \bar{\rho} s_t - q/\beta}{\gamma \sigma^2 (1 - \bar{\rho}^2)}, & s_t \leq -\lambda \bar{\rho}, \\ n^0, & -\lambda \bar{\rho} \leq s_t \leq -\lambda \underline{\rho}, \\ \frac{\mu + \sigma \underline{\rho} s_t - q/\beta}{\gamma \sigma^2 (1 - \underline{\rho}^2)}, & s_t \geq -\lambda \underline{\rho}. \end{cases} \quad (30)$$

The three pieces paste together continuously. The policy is flat on the inertia region, and outside the region its slope is

$$\frac{\partial n^*(s_t; \hat{\rho})}{\partial s_t} = \frac{\hat{\rho}}{\gamma \sigma (1 - \hat{\rho}^2)},$$

with $\hat{\rho} = \bar{\rho}$ on the bad-news side and $\hat{\rho} = \underline{\rho}$ on the good-news side.

The three pieces meet continuously but with kinks. At $s_t = -\lambda \bar{\rho}$, the bad-news branch reaches n^0 ; at $s_t = -\lambda \underline{\rho}$, the good-news branch does the same. Because $\rho/(1 - \rho^2)$ rises on $[0, 1)$, fertility reacts more sharply on the bad-news branch, governed by $\bar{\rho}$, than on the good-news branch, governed by $\underline{\rho}$. This is the model's asymmetry in its clearest form: parents act strongly on sufficiently adverse news while discounting favorable news.

4.6 Generosity and legibility of family policy

Up to this point, policy has appeared only as part of the public signal. I now introduce a family benefit directly. The central result is that a household responds to what it can safely expect to receive, which need not bear much relation to the program's headline cost.

Consider a statutory subsidy $\tau \geq 0$ for each child. Its announced value is not necessarily the amount a household expects to collect over the years in which it will raise that child. Eligibility may be means-tested, childcare places may be rationed, applications may be burdensome, or a later government may change the program. Let α denote the fraction of the statutory benefit that ultimately reaches the household. Rather than assigning a prior to α , the household can only place it in the interval

$$\alpha \in [\underline{\alpha}, \bar{\alpha}] \subseteq [0, 1], \quad (31)$$

just as it does with signal informativeness. In this notation, a program is more legible when its lower bound $\underline{\alpha}$ is higher and the range $[\underline{\alpha}, \bar{\alpha}]$ is tighter. A clearer information environment likewise has a narrower $[\underline{\rho}, \bar{\rho}]$. Because the net cost of a child is $q - \alpha\tau$, the closed-form parent in Section 4.5 solves

$$\max_{0 \leq n \leq y/(q - \alpha\tau)} \min_{\substack{\rho \in [\underline{\rho}, \bar{\rho}] \\ \alpha \in [\underline{\alpha}, \bar{\alpha}]}} \left\{ y - (q - \alpha\tau)n + \beta \left[n(\mu + \sigma \rho s_t) - \frac{\gamma}{2} n^2 \sigma^2 (1 - \rho^2) \right] \right\}. \quad (32)$$

For any $n > 0$ and $\tau > 0$, the objective rises with α . The least-favorable evaluation therefore sets $\alpha = \underline{\alpha}$, regardless of the family plan or the value chosen for ρ . The two minimizations separate, and the earlier results carry over after replacing q with

$$q_e(\tau) \equiv q - \underline{\alpha}\tau.$$

Proposition 5 (Generosity and legibility). *Consider problem (32) with $0 \leq \underline{\alpha} \leq \bar{\alpha} \leq 1$ and $q - \underline{\alpha}\tau > 0$, and suppose the signal-free solution is interior. Then the following hold.*

- (i) *The worst-case pass-through is $\alpha^* = \underline{\alpha}$ for every plan with $n > 0$, and all results of Sections 4.1–4.5 hold with q replaced by $q_e(\tau)$. In particular, inside the inertia region fertility equals*

$$n^0(\tau) = \frac{\mu - (q - \underline{\alpha}\tau)/\beta}{\gamma\sigma^2}. \quad (33)$$

- (ii) *The fertility response to the subsidy is*

$$\frac{\partial n^0}{\partial \tau} = \frac{\underline{\alpha}}{\beta\gamma\sigma^2}. \quad (34)$$

The response is scaled by the worst-case pass-through $\underline{\alpha}$, not by the statutory size τ .

- (iii) *(Legibility equivalence.) An ambiguous program $(\tau, [\underline{\alpha}, \bar{\alpha}])$ has the same effect on every household decision as a fully legible program $(\alpha \equiv 1, \text{known})$ of size $\underline{\alpha}\tau$. Equivalently, for $\underline{\alpha} > 0$, replicating the effect of a legible program of size τ^* requires scaling the ambiguous program up by the factor $1/\underline{\alpha}$. For $\underline{\alpha} = 0$, no finite statutory subsidy replicates a positive legible one.*

- (iv) *The width of the inertia region, $\lambda(\tau)(\bar{\rho} - \underline{\rho})$ with $\lambda(\tau) = [\mu - (q - \underline{\alpha}\tau)/\beta]/\sigma$, is nondecreasing in τ ,*

$$\frac{\partial}{\partial \tau} [\lambda(\tau)(\bar{\rho} - \underline{\rho})] = \frac{\underline{\alpha}}{\beta\sigma}(\bar{\rho} - \underline{\rho}) \geq 0,$$

with strict inequality if and only if $\underline{\alpha} > 0$ and $\bar{\rho} > \underline{\rho}$.

Proof. The proof is available in Appendix A.6. □

Part (ii) includes a stark limiting case. As $\underline{\alpha} \rightarrow 0$, the fertility response vanishes for any statutory benefit, even though government spending need not. If the realized pass-through is some $\alpha_0 \in [\underline{\alpha}, \bar{\alpha}]$ with $\alpha_0 > 0$, the government pays $g = \alpha_0\tau$ per child and $g = \alpha_0\tau$, and the fertility response per unit of realized per-child transfer is

$$\frac{\partial n^0}{\partial g} = \frac{\underline{\alpha}}{\alpha_0} \cdot \frac{1}{\beta\gamma\sigma^2} \leq \frac{1}{\beta\gamma\sigma^2},$$

which is the response to a fully legible transfer multiplied by $\underline{\alpha}/\alpha_0 \leq 1$. Put plainly, the household plans against the lower bound, while the treasury pays according to the realized program.

This wedge offers a formal account of the motivating empirical pattern. A program can be expensive and still have little effect on births if families cannot rule out receiving only a small share of it. The relevant gap is between realized pass-through, α_0 , and the lower bound that guides the household's decision, $\underline{\alpha}$. Annual renewals, scheduled expirations, means-tested phase-outs, and lotteries for scarce places can all reduce that lower bound without changing the program's expected value. The South Korean baby bonus provides a useful scale comparison. The benefit was financed locally and renewed each year by district councils; Kim (2024) estimates elasticities of only 0.03 to 0.06 and calculates that replacement fertility would require a transfer ten to sixteen times larger, roughly the median household's annual income. In the model, such a

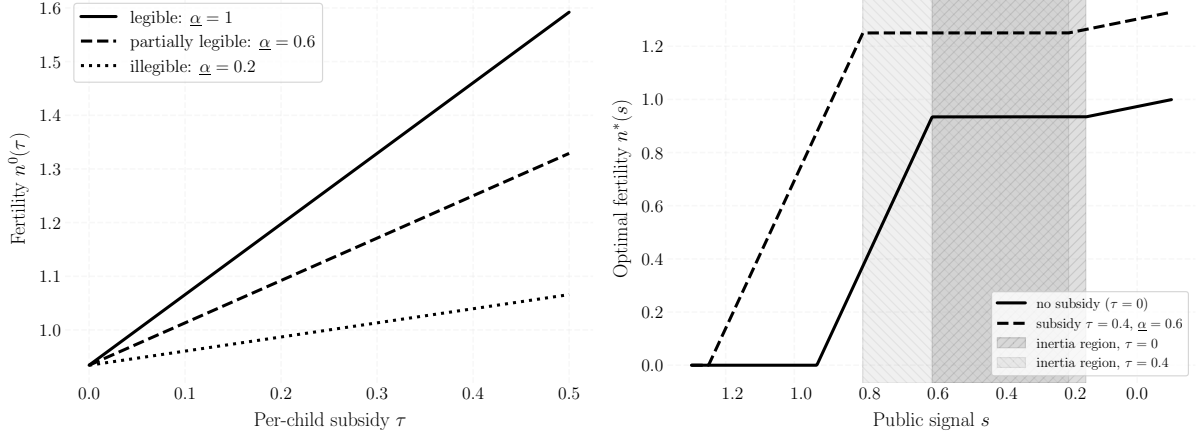


Figure 2: Generosity versus legibility of family policy

Notes: The left panel plots inertia-level fertility $n^0(\tau)$ in (33) against the statutory per-child subsidy τ for three levels of worst-case pass-through $\underline{\alpha}$. The slope is $\underline{\alpha}/(\beta\gamma\sigma^2)$, so the same statutory generosity produces responses ranked by legibility. The right panel plots the full policy function $n^*(s)$ without a subsidy and with a subsidy $\tau = 0.4$ under $\underline{\alpha} = 0.6$. The subsidy raises the inertia level but also shifts and widens the inertia region (Proposition 5(iv)). Parameter values are $u(c) = c$, $\Phi(n) = n$, $y = 10$, $q = 1$, $\beta = 0.95$, $\mu = 1.8$, $\sigma = 1$, $\gamma = 0.8$, $\underline{\rho} = 0.2$, and $\bar{\rho} = 0.8$.

small response is what one obtains when $\underline{\alpha}$ lies well below one: each promised dollar shifts the household's decision by only a fraction of the benchmark amount.

Part (iv) points to a second, less obvious effect. A subsidy raises the value at stake in the parent's dynastic position. The parent then becomes cautious over a wider range of news, so the inertia interval expands. Spending more through a program that families do not trust can therefore make the response to the next announcement even weaker.

The announcement itself is also a public signal. A credible new program is good news, but Corollary 2 says that the parent reads good news using $\underline{\rho}$. Its effect on fertility therefore has the muted slope $\underline{\rho}/[\gamma\sigma(1 - \underline{\rho}^2)]$. In practice, governments often announce family packages after demographic or economic conditions have deteriorated. The package and the circumstances that prompted it may together fall inside $[-\lambda\bar{\rho}, -\lambda\underline{\rho}]$. Fertility then does not move at all, even though households have paid attention. An event-study coefficient near zero need not imply that the announcement went unnoticed.

The natural policy response is to make the program easier to read and rely on. Such a reform raises $\underline{\alpha}$ toward $\bar{\alpha}$, because a household can verify what it will receive, and narrows $[\underline{\rho}, \bar{\rho}]$, because public conditions become easier to map into the household's own future. These changes affect the inertia width $W = \lambda(\tau, \underline{\alpha})(\bar{\rho} - \underline{\rho})$ through different channels. Raising $\underline{\alpha}$ improves the pass-through of each dollar already on offer. When $\tau > 0$, however, it also raises the dynastic value at stake and, by the force in Proposition 5(iv), tends to widen W : $\partial W/\partial \underline{\alpha} = \tau(\bar{\rho} - \underline{\rho})/(\beta\sigma) \geq 0$. By contrast, narrowing the informativeness interval unambiguously reduces the width: $\partial W/\partial(\bar{\rho} - \underline{\rho}) = \lambda > 0$. The total effect,

$$dW = \frac{\tau(\bar{\rho} - \underline{\rho})}{\beta\sigma} d\underline{\alpha} + \lambda d(\bar{\rho} - \underline{\rho}),$$

The inertia region shrinks only when the reduction in $\bar{\rho} - \underline{\rho}$ outweighs the exposure effect of the higher pass-through. Thus legibility always increases the return to a given subsidy, but it restores responsiveness to news only if its informational component is large enough. Figure 2 shows the pass-through channel. Guaranteed rather than rationed childcare, statutory eligibility, enforceable return-to-work rights, and benefit schedules fixed for a credible horizon all operate on the bounds themselves, not just on τ . They make the promised payoff easier to verify and more likely to arrive. The model therefore predicts that a specific, dependable benefit can move

births at a lower fiscal cost than a larger but poorly defined promise.

4.7 How special is exact flatness?

Exact flatness in the benchmark relies on maxmin preferences and a continuous fertility choice. I consider two departures—smooth ambiguity and integer fertility—to see which parts of the mechanism remain. These are targeted robustness exercises, not a claim that ambiguity is necessary for every form of inaction. Appendix B gives the details.

Under smooth ambiguity (Klibanoff et al., 2005), the parent has a prior π over $[\underline{\rho}, \bar{\rho}]$ and an ambiguity-aversion parameter η . As η grows, the value criterion approaches maxmin, and the endogenous tilt over ρ points in the same direction as the worst-case selection: worse news places more weight on less favorable interpretations (Appendix B.1). Exact flatness appears only in the limit $\eta = \infty$. With finite η , the policy is smooth, and its slope contains a covariance term whose sign is not fixed. One therefore cannot assert monotonicity or a universal S-shape without more structure. Convergence of the policy itself, as opposed to its value, also requires the additional conditions stated in the appendix.

Now suppose children come only in whole numbers (Proposition B.2.1). The parent compares neighboring plans, such as n and $n + 1$. If the worst-case belief is interior for both, their values contain the same penalty, $-s_t^2/(2\gamma)$, which cancels from the comparison. Bad news lowers welfare without changing which family size is preferred. Except at an exact tie, the parent cannot switch plans anywhere in this common interior region. Switching occurs only after a worst-case belief reaches a boundary; $\bar{\rho}$ then governs the bad-news side and $\underline{\rho}$ the good-news side.

This distinction should be visible in the data. Survey answers about intentions or economic conditions may deteriorate after bad news, while realized births remain unchanged, as in Section 2. Integer choice by itself creates intervals of inaction, so I do not claim that ambiguity always makes those intervals wider. Relative to a known- ρ model, their width can increase or decrease depending on the comparison value of ρ . The robust implication is instead the placement of the switching points and the separation between welfare and behavior while both plans have interior worst-case beliefs.

5 Conclusion

Why can a government spend heavily on families and see so little change in births? The model in this paper suggests that the missing margin is legibility. I introduce uncertainty about signal informativeness into the Barro–Becker framework (Becker and Barro, 1988; Barro and Becker, 1989). A parent who sees public news does not know exactly how it maps into the future of her own children and evaluates each plan using the least favorable admissible interpretation.

That simple change produces fertility inertia. Good news is discounted, very bad news is taken seriously, and moderately bad news changes welfare without changing fertility, consumption, or investment in children. The flat response is exact under maxmin preferences. Smooth ambiguity preserves the pessimistic tilt but generally smooths the policy, while integer fertility preserves an exact separation between welfare and behavior over the region in which neighboring plans share an interior worst-case belief.

The same logic applies to a family program. What matters for the household is not the benefit printed in the statute, but the amount it can count on receiving. If that lower bound is only a fraction of the headline benefit, an opaque program must be correspondingly larger to match the effect of a clear entitlement. The treasury pays what the program ultimately delivers; families act on what they can safely expect. This is why generosity alone is an incomplete measure of policy. Eligibility that can be verified, assured childcare places, enforceable leave and return-to-work rights, and commitments that survive the relevant planning horizon can make

existing spending more effective. They change fertility incentives by making the promise itself credible.

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Materials availability Not applicable.

Code availability Figure-generation and replication code is available from the author upon request.

Author contribution HongSeok Kim is the sole author and completed all aspects of the work.

Appendices

A Proofs

A.1 Derivation of CARA

Given the future net payoff of a marginal child X_{t+1} , the conditional mean

$$\mathbb{E}[X_{t+1}|s, \rho] = \mu + \sigma\rho s$$

and

$$\text{Var}(X_{t+1} | s, \rho) = \sigma^2(1 - \rho^2).$$

The total payoff of the parent having n children is

$$Z_{t+1} = \Phi(n)X_{t+1}$$

where $\Phi(n)$ is the effective dynastic discounting factor. Conditional on signal s and parameter ρ ,

$$Z_{t+1}|s, \rho \sim N(m, v)$$

where

$$m = \Phi(n)[\mu + \sigma\rho s]$$

and

$$v = \Phi(n)^2\sigma^2(1 - \rho^2).$$

Given that the utility function is CARA, the parent evaluates the risky future dynastic payoff Z ,

$$U(Z) = -\exp(-\gamma Z)$$

where $\gamma > 0$ is absolute risk aversion. Let

$$-\exp(-\gamma CE) = \mathbb{E}(-\exp(-\gamma Z))$$

and using $Z \sim N(m, v)$, we get

$$\mathbb{E}[\exp(-\gamma Z)] = \exp\left(-\gamma m + \frac{1}{2}\gamma^2 v\right).$$

Therefore,

$$\exp(-\gamma CE) = \exp\left(-\gamma m + \frac{1}{2}\gamma^2 v\right)$$

which gives

$$CE = m - \frac{\gamma}{2}v = \mathbb{E}[Z] - \frac{\gamma}{2}\text{Var}(Z).$$

Substituting the conditional mean and variance of random variable Z yields the equation

$$CE(n; s, \rho) = \Phi(n)[\mu + \sigma\rho s] - \frac{\gamma}{2}\Phi(n)^2\sigma^2(1 - \rho^2).$$

A.2 Proof of Proposition 1

Proof. Let $C_b(\mathcal{X} \times \mathcal{S})$ be the space of bounded continuous functions on the state space, with the sup norm $\|\cdot\|_\infty$, a complete metric space. Write a feasible action as $a = (c, n, z) \in \Gamma(x, s)$ and $V' = V(x', s')$. We show T maps C_b into itself and is a contraction with modulus $\beta\bar{\Phi}$, then characterize the fixed point by measurable selections.

We use two properties of $\mathcal{C}_{\gamma, \rho, z}$, both immediate from $\mathcal{C}_{\gamma, \rho, z}[Y] = -\gamma^{-1} \log \mathbb{E}_{\rho, z}[e^{-\gamma Y}]$ (and from the conditional expectation when $\gamma = 0$). The properties are monotonicity, $Y_1 \leq Y_2 \Rightarrow \mathcal{C}_{\gamma, \rho, z}[Y_1] \leq \mathcal{C}_{\gamma, \rho, z}[Y_2]$, and translation invariance, $\mathcal{C}_{\gamma, \rho, z}[Y + k] = \mathcal{C}_{\gamma, \rho, z}[Y] + k$ for constant k . Together they give the sup-norm bound. If $\|Y_1 - Y_2\|_\infty \leq \eta$ then

$$|\mathcal{C}_{\gamma, \rho, z}[Y_1] - \mathcal{C}_{\gamma, \rho, z}[Y_2]| \leq \eta. \quad (*)$$

T maps C_b into C_b . Fix $V \in C_b$. By Assumption 1, Γ is nonempty, compact-valued, and continuous, $[\rho, \bar{\rho}]$ is compact, u is continuous and bounded on the graph of Γ ($|u| \leq \bar{u}$), and $0 \leq \Phi(n) \leq \bar{\Phi}$. Then $|\Phi(n)V'| \leq \bar{\Phi}\|V\|_\infty$, and applying $(*)$ against the constant bounds gives $|\mathcal{C}_{\gamma, \rho, z}[\Phi(n)V']| \leq \bar{\Phi}\|V\|_\infty$ uniformly in ρ , hence $|(TV)(x, s)| \leq \bar{u} + \beta\bar{\Phi}\|V\|_\infty$, so TV is bounded. For continuity, set $H_V(x, s, a, \rho) = u(x, s, a) + \beta\mathcal{C}_{\gamma, \rho, z}[\Phi(n)V']$. Under Assumption 1 the transition law is continuous and the integrand $e^{-\gamma\Phi(n)V'}$ is bounded and continuous, so by bounded convergence H_V is continuous in (x, s, a, ρ) . Since $[\rho, \bar{\rho}]$ is compact and fixed in (x, s, a) , Berge's theorem makes $G_V(x, s, a) := \min_\rho H_V(x, s, a, \rho)$ continuous with nonempty compact-valued argmin. Applying Berge's theorem again with the compact-valued continuous Γ makes $(TV)(x, s) = \max_{a \in \Gamma(x, s)} G_V(x, s, a)$ continuous.

T is a contraction. Let $V_1, V_2 \in C_b$ and $\Delta := \|V_1 - V_2\|_\infty$. For fixed feasible (x, s, a, ρ) , $|V_1' - V_2'| \leq \Delta$ and $\Phi(n) \geq 0$ give $\|\Phi(n)V_1' - \Phi(n)V_2'\|_\infty \leq \bar{\Phi}\Delta$, so by $(*)$,

$$|\mathcal{C}_{\gamma, \rho, z}[\Phi(n)V_1'] - \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V_2']| \leq \bar{\Phi}\Delta$$

uniformly in ρ . The minimization over ρ and the maximization over a are each nonexpansive in the sup norm. If $\sup_\rho |f_\rho - g_\rho| \leq \eta$ then $|\min_\rho f_\rho - \min_\rho g_\rho| \leq \eta$, and likewise for $\max$ (from $f \leq g + \eta$, take the relevant extremum of both sides, then swap roles). The payoff $u(x, s, a)$ cancels in the difference. Passing the bound through the inner min, multiplying by β , and passing through the outer max gives, for every (x, s) ,

$$|(TV_1)(x, s) - (TV_2)(x, s)| \leq \beta\bar{\Phi}\Delta,$$

hence $\|TV_1 - TV_2\|_\infty \leq \beta\bar{\Phi}\|V_1 - V_2\|_\infty$. Since $\beta\bar{\Phi} < 1$ by Assumption 1, T is a contraction on C_b , and Banach's fixed-point theorem gives a unique $V \in C_b$ with $V = TV$.

Characterization. At this fixed point define

$$\begin{aligned} \mathcal{R}_V(x, s, a) &:= \arg \min_{\rho \in [\rho, \bar{\rho}]} \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V'], \\ \mathcal{P}_V(x, s) &:= \arg \max_{a \in \Gamma(x, s)} \left\{ u(x, s, a) + \beta \min_{\rho} \mathcal{C}_{\gamma, \rho, z}[\Phi(n)V'] \right\}. \end{aligned}$$

By the continuity and compactness above, Berge's theorem makes both correspondences nonempty and compact-valued, each admitting a measurable selection. For any selections $a^*(x, s) = (c^*, n^*, z^*)(x, s) \in \mathcal{P}_V(x, s)$ and $\rho^*(x, s) \in \mathcal{R}_V(x, s, a^*(x, s))$,

$$V(x, s) = u(x, s, a^*(x, s)) + \beta \mathcal{C}_{\gamma, \rho^*(x, s), z^*(x, s)}[\Phi(n^*(x, s))V'] \quad \text{for every } (x, s).$$

The worst-case belief $\rho^*(x, s)$ is the inner minimizer evaluated at the chosen policy $a^*(x, s)$, which is the belief endogeneity used in the inertia analysis. $\square$

A.3 Proof of Proposition 2

Proof. Fix the current state (x_t, s_t) and suppress it where no confusion can arise. Let

$$J(n, z; x_t, s_t) := \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n, z; x_t, s_t).$$

The parent's problem can be written as

$$\max_{c, n, z} \{u(c) + \beta J(n, z; x_t, s_t)\}$$

subject to

$$c + n(q_t + z) \leq m_t.$$

Since $u'(c) > 0$, the budget constraint binds at an optimum. Hence

$$c_t^* = m_t - n_t^*(q_t + z_t^*),$$

which gives (12).

Let λ_t denote the multiplier on the budget constraint. For an interior solution, the first-order conditions are

$$u'(c_t^*) = \lambda_t,$$

$$\beta J_n(n_t^*, z_t^*; x_t, s_t) = \lambda_t(q_t + z_t^*),$$

and

$$\beta J_z(n_t^*, z_t^*; x_t, s_t) = \lambda_t n_t^*.$$

Substituting $\lambda_t = u'(c_t^*)$ gives

$$u'(c_t^*)(q_t + z_t^*) = \beta J_n(n_t^*, z_t^*; x_t, s_t)$$

and

$$n_t^* u'(c_t^*) = \beta J_z(n_t^*, z_t^*; x_t, s_t).$$

It remains to compute the derivatives of J . For fixed ρ , use the differentiable shock representation

$$x_{t+1} = G^x(x_t, s_t, z_t, \varepsilon_{t+1}; \rho), \quad s_{t+1} = G^s(x_t, s_t, z_t, \varepsilon_{t+1}; \rho),$$

where the conditional law of ε_{t+1} depends on ρ but not on n or z (Assumption 1(v)). This common-shock structure is what keeps the derivative below free of a likelihood-score term. If instead the law of the shock varied directly with z , with density $p_{\rho, z}$, the investment derivative would contain the additional term $-\gamma^{-1} \mathbb{E}_Q[\partial_z \log p_{\rho, z}]$. Define

$$Y_{t+1}(n, z, \rho) := \Phi(n) V(G^x(x_t, s_t, z, \varepsilon_{t+1}; \rho), G^s(x_t, s_t, z, \varepsilon_{t+1}; \rho)).$$

For $\gamma > 0$,

$$K_\rho(n, z; x_t, s_t) = -\frac{1}{\gamma} \log \mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}(n, z, \rho)\} \mid x_t, s_t],$$

where the expectation is over the shock ε_{t+1} .

By the maintained differentiability and domination assumptions, differentiation under the expectation is valid. For any control $a \in \{n, z\}$,

$$\frac{\partial K_\rho}{\partial a} = -\frac{1}{\gamma} \frac{\mathbb{E}_\rho \left[-\gamma \exp\{-\gamma Y_{t+1}\} \frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t \right]}{\mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t]} = \frac{\mathbb{E}_\rho \left[\exp\{-\gamma Y_{t+1}\} \frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t \right]}{\mathbb{E}_\rho[\exp\{-\gamma Y_{t+1}\} \mid x_t, s_t]}.$$

Define the distorted measure $Q_{\rho,n,z}$ by

$$\frac{dQ_{\rho,n,z}}{dP_{\rho,z}} = \frac{\exp\{-\gamma Y_{t+1}(n, z, \rho)\}}{\mathbb{E}_{\rho,z}[\exp\{-\gamma Y_{t+1}(n, z, \rho)\} \mid x_t, s_t]}.$$

Equivalently, under the shock representation, this is the exponential tilt of the shock law. Therefore

$$\frac{\partial K_\rho}{\partial a} = \mathbb{E}_{Q_{\rho,n,z}} \left[\frac{\partial Y_{t+1}}{\partial a} \mid x_t, s_t \right].$$

Since the inner minimizer ρ_t^* is unique, Danskin's envelope theorem (Danskin, 1966; Milgrom and Segal, 2002) implies

$$J_a(n_t^*, z_t^*; x_t, s_t) = \left. \frac{\partial K_\rho(n_t^*, z_t^*; x_t, s_t)}{\partial a} \right|_{\rho=\rho_t^*}, \quad a \in \{n, z\}.$$

Thus there is no additional term involving the derivative of ρ_t^* with respect to n or z .

Now compute the two derivatives of Y_{t+1} . Since the transition does not depend directly on n ,

$$\frac{\partial Y_{t+1}}{\partial n} = \Phi'(n_t^*) V(x_{t+1}, s_{t+1}).$$

Therefore

$$J_n(n_t^*, z_t^*; x_t, s_t) = \Phi'(n_t^*) \mathbb{E}_{Q_t^*} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t],$$

where $Q_t^* = Q_{\rho_t^*, n_t^*, z_t^*}$. Substituting into the first-order condition for n gives

$$u'(c_t^*)(q_t + z_t^*) = \beta \Phi'(n_t^*) \mathbb{E}_{Q_t^*} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t],$$

which is (13).

For z , the chain rule gives

$$\begin{aligned} \frac{\partial Y_{t+1}}{\partial z} &= \Phi(n_t^*) [\nabla_x V(x_{t+1}, s_{t+1}) \cdot G_z^x(x_t, s_t, z_t^*, \varepsilon_{t+1}; \rho_t^*) \\ &\quad + V_s(x_{t+1}, s_{t+1}) G_z^s(x_t, s_t, z_t^*, \varepsilon_{t+1}; \rho_t^*)] \\ &= \Phi(n_t^*) D_z V_{t+1}. \end{aligned}$$

Hence

$$J_z(n_t^*, z_t^*; x_t, s_t) = \Phi(n_t^*) \mathbb{E}_{Q_t^*} [D_z V_{t+1} \mid x_t, s_t].$$

Substituting into the first-order condition for z gives

$$n_t^* u'(c_t^*) = \beta \Phi(n_t^*) \mathbb{E}_{Q_t^*} [D_z V_{t+1} \mid x_t, s_t],$$

which is (14).

Finally, by definition of the inner problem,

$$\rho_t^* \in \arg \min_{\rho \in [\underline{\rho}, \bar{\rho}]} K_\rho(n_t^*, z_t^*; x_t, s_t).$$

If the minimizer is interior and K_ρ is differentiable in ρ , the usual first-order condition gives $\partial_\rho K_\rho|_{\rho=\rho_t^*} = 0$. If $\rho_t^* = \underline{\rho}$ the right derivative must satisfy $\partial_\rho^+ K_\rho|_{\rho=\underline{\rho}} \geq 0$, and if $\rho_t^* = \bar{\rho}$ the left derivative must satisfy $\partial_\rho^- K_\rho|_{\rho=\bar{\rho}} \leq 0$, which is (15). This proves the result. $\square$

A.4 Proof of Proposition 3

Proof. Fix $z_t = z$ and write

$$c(n) = m_t - n(q_t + z).$$

The fertility first-order condition is

$$G(n, \rho; s_t) \equiv -(q_t + z)u'(c(n)) + \beta\Phi'(n) \left[\mu(z) + \sigma(z)\rho s_t - \gamma\Phi(n)\sigma(z)^2(1 - \rho^2) \right] = 0.$$

At the interior optimum, the second-order condition implies $G_n(n^*, \rho; s_t) < 0$. Hence the implicit function theorem gives

$$\frac{\partial n^*}{\partial \rho} = -\frac{G_\rho}{G_n}.$$

Differentiating G with respect to ρ gives

$$G_\rho = \beta\Phi'(n^*)\sigma(z) [s_t + 2\gamma\Phi(n^*)\sigma(z)\rho].$$

Since $\beta > 0$, $\Phi'(n^*) > 0$, $\sigma(z) > 0$, and $G_n < 0$, the sign of $\partial n^*/\partial \rho$ is the sign of

$$s_t + 2\gamma\Phi(n^*)\sigma(z)\rho,$$

which proves (20).

For the case with endogenous z , let

$$W(n, z; \rho, s_t) = u(m_t - n(q_t + z)) + \beta K(n, z; \rho, s_t).$$

The first-order conditions are

$$W_n(n^*, z^*; \rho, s_t) = 0, \quad W_z(n^*, z^*; \rho, s_t) = 0.$$

Totally differentiating this system with respect to ρ yields

$$H \begin{pmatrix} \partial n^*/\partial \rho \\ \partial z^*/\partial \rho \end{pmatrix} + \begin{pmatrix} W_{n\rho} \\ W_{z\rho} \end{pmatrix} = 0,$$

where H is the Hessian of W with respect to (n, z) . Direct differentiation gives

$$W_{n\rho} = \beta\Phi'(n^*)\sigma(z^*) [s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho],$$

and

$$W_{z\rho} = \beta\Phi(n^*)\sigma_z(z^*) [s_t + 2\gamma\Phi(n^*)\sigma(z^*)\rho].$$

Solving the linear system gives (21). □

A.5 Proof of Proposition 4

Proof. Wherever the worst-case belief is interior, (22) gives

$$\rho^w(n, z, s_t) = -\frac{s_t}{\gamma\Phi(n)\sigma(z)}.$$

Substituting this value into (18) gives the minimized continuation value (26),

$$\min_{\rho} K(n, z; \rho, s_t) = \Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma}.$$

The last term depends on the signal, but it does not depend on either choice variable, n or z .

Therefore, on any region of the choice space where the worst-case belief is interior, the parent's problem is the signal-free problem (23) plus the constant $-\beta s_t^2/(2\gamma)$. Adding a constant changes the level of value but not the maximizer.

The signal restriction (24) is equivalent to

$$\underline{\rho} < -\frac{s_t}{\gamma\Phi(n^0)\sigma(z^0)} < \bar{\rho}.$$

Thus the worst-case belief is interior at (n^0, z^0) . By continuity, it remains interior in a neighborhood of (n^0, z^0) . On that neighborhood, the objective is exactly the signal-free objective plus a constant. Since (n^0, z^0) is the unique local maximizer of the signal-free problem, it is also the unique local maximizer of the ambiguous-signal problem. Hence

$$n^*(s_t) = n^0, \quad z^*(s_t) = z^0,$$

and the budget condition gives

$$c^*(s_t) = m_t - n^0(q_t + z^0).$$

The constant-shift argument extends globally only across plans whose worst-case belief is interior. For a plan whose worst case sits at a boundary $\hat{\rho} \in \{\underline{\rho}, \bar{\rho}\}$, completing the square in (18) gives

$$\min_{\rho \in [\underline{\rho}, \bar{\rho}]} K = \Phi(n)\mu(z) - \frac{\gamma}{2}\Phi(n)^2\sigma(z)^2 - \frac{s_t^2}{2\gamma} + \frac{\gamma}{2}\left[\Phi(n)\sigma(z)\hat{\rho} + \frac{s_t}{\gamma}\right]^2,$$

whose last term is plan-dependent and nonnegative, so such plans move with the signal relative to (n^0, z^0) . If the worst-case belief is interior at every candidate optimizer, the signal-free objective is globally strictly concave on that set, and (n^0, z^0) weakly dominates all boundary-belief plans and $n = 0$, then (25) is the unique global optimum.

The final step checks the endpoints of (24). At an endpoint the worst-case belief at (n^0, z^0) no longer lies strictly inside the ambiguity interval. It sits exactly at one boundary $b \in \{\underline{\rho}, \bar{\rho}\}$, so (n^0, z^0) is the knife-edge case between the interior regime and the boundary regime. Write $A(n, z) = \Phi(n)\sigma(z)$ for the exposure of a plan and $A^0 = A(n^0, z^0)$. At the endpoint the signal satisfies $s_t = -\gamma b A^0$. Substituting this into the boundary formula above shows that the minimized continuation value on the boundary side equals its interior-side counterpart plus the extra term $\frac{\gamma b^2}{2}[A(n, z) - A^0]^2$. The extra term vanishes at (n^0, z^0) and is positive at any plan with a different exposure. It is a convex reward for moving away from (n^0, z^0) , so it pushes against the concavity of the objective, and whether (n^0, z^0) survives as a local maximum is a second-order comparison. Let H_0 be the Hessian of the interior-regime objective at (n^0, z^0) , which is negative definite by assumed strict concavity. After discounting by β , the extra term adds the Hessian $\beta\gamma b^2 \nabla A \nabla A'$, a positive semidefinite matrix of rank one. The plan (n^0, z^0) remains a strict local maximum at the endpoint exactly when concavity wins, that is, when $H_0 + \beta\gamma b^2 \nabla A \nabla A' \prec 0$. This is the strengthened second-order condition referenced in the proposition. In the one-dimensional closed form, with $\Phi(n) = n$ and z fixed, both matrices are scalars. There $H_0 = -\beta\gamma\sigma^2$ and the extra term adds $\beta\gamma b^2\sigma^2$, so the condition reads $-\beta\gamma\sigma^2(1 - b^2) < 0$. It holds automatically because $b^2 < 1$. $\square$

A.6 Proof of Proposition 5

Proof. Write the inner objective of (32) as

$$W(n; \rho, \alpha, s_t) = y - (q - \alpha\tau)n + \beta \left[n(\mu + \sigma\rho s_t) - \frac{\gamma}{2}n^2\sigma^2(1 - \rho^2) \right].$$

Part (i). For any (n, ρ) with $n > 0$ and $\tau > 0$,

$$\frac{\partial W}{\partial \alpha} = \tau n \geq 0,$$

so W is nondecreasing in α and the inner minimum over $\alpha \in [\underline{\alpha}, \bar{\alpha}]$ is attained at $\alpha^* = \underline{\alpha}$ for every plan and every ρ . (If $n = 0$ or $\tau = 0$, the objective does not depend on α and the selection $\alpha^* = \underline{\alpha}$ is without loss.) Because α enters W only through the additive term $\alpha\tau n$, which does not depend on ρ , the joint minimization over (ρ, α) separates,

$$\min_{\rho, \alpha} W(n; \rho, \alpha, s_t) = \min_{\rho \in [\underline{\rho}, \bar{\rho}]} W(n; \rho, \underline{\alpha}, s_t).$$

The resulting problem is exactly the closed-form problem of Section 4.5 with the child cost $q_e(\tau) = q - \underline{\alpha}\tau$ in place of q . All results of Sections 4.1–4.5 therefore apply verbatim with this substitution. In particular, (28) becomes (33).

Part (ii). Differentiating (33) with respect to τ ,

$$\frac{\partial n^0}{\partial \tau} = \frac{\underline{\alpha}}{\beta\gamma\sigma^2}.$$

Part (iii). By Part (i), the household's objective under the ambiguous program $(\tau, [\underline{\alpha}, \bar{\alpha}])$ coincides, for every plan n and every signal s_t , with the objective under a known pass-through $\alpha \equiv 1$ and statutory size $\underline{\alpha}\tau$, since both produce the same effective cost $q - \underline{\alpha}\tau$. Identical objectives imply identical argmax correspondences, hence identical policies. Matching a legible program of size τ^* requires $\underline{\alpha}\tau = \tau^*$, i.e. $\tau = \tau^*/\underline{\alpha}$.

Part (iv). By Part (i) and the definition of λ in Section 4.5,

$$\lambda(\tau) = \frac{\mu - (q - \underline{\alpha}\tau)/\beta}{\sigma}, \quad \frac{\partial \lambda}{\partial \tau} = \frac{\underline{\alpha}}{\beta\sigma} > 0$$

whenever $\underline{\alpha} > 0$, and the width of the inertia region is $\lambda(\tau)(\bar{\rho} - \underline{\rho})$ by (29). The illegible-program limit discussed after Proposition 5 follows by taking $\underline{\alpha} \rightarrow 0$ in Part (ii) and noting that the realized outlay per household is $\alpha_0\tau n^0 > 0$ whenever $\alpha_0 > 0$ and $n^0 > 0$. $\square$

A.7 Dynamic consistency of the composed criterion

This appendix expands on the discussion following Assumption 2. Dynamic consistency asks whether the parent would ever want to abandon a plan midway. A criterion is dynamically consistent when the plan that looks best at date t , viewed from the whole future, still looks best at date $t + 1$ once the recursion is rolled forward one step. The criterion in (6) stacks two layers on top of the expectation, a worst case over ρ and a CARA certainty equivalent, and each layer raises this question separately.

The maxmin layer is the familiar case. Worst-case criteria are vulnerable to a specific failure. If nature must commit to one ρ for all time, the parent at date 0 guards against a single worst ρ for the whole future, while the parent at date 1 re-solves against a possibly different worst ρ , and the two plans can disagree. Epstein and Schneider (2003) show that this failure disappears when the set of priors over paths is rectangular, meaning that nature can pick the worst one-step-ahead law afresh at every date and history. Assumption 2 imposes exactly this structure. The path-prior set is built from arbitrary history-by-history selections $\rho_t(h_t) \in [\underline{\rho}, \bar{\rho}]$, so whatever conditional law the date 1 parent fears was already available to nature in the date 0 parent's problem. The two problems therefore share the same worst case, and the maxmin layer is dynamically consistent by the Epstein–Schneider theorem.

The CARA layer requires a different argument. The certainty equivalent $-\frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma Y}]$ is a nonlinear transformation of the expectation, and the Epstein–Schneider theorem covers only the

maximin layer, not a nonlinear layer composed with it. No general theorem delivers consistency of the composition. We therefore follow the standard route in recursive robust control (Hansen and Sargent, 2008) and define preferences directly by the recursion (6). On this reading the parent does not first form a criterion over entire future paths and then check that the recursion reproduces it. The recursion itself is the primitive. Each generation evaluates its choice by current utility plus the composed adjustment of the next generation's value, and the question of a conflicting path-based evaluation never arises because no path-based evaluation is defined. Dynamic consistency holds by construction.

The entropy dual of Appendix A.8 gives the same conclusion a game interpretation. Each period the parent moves, and nature responds twice, once by selecting ρ_t from the interval and once by distorting the conditional law subject to an entropy penalty scaled by $1/\gamma$. Both of nature's moves are one-step-ahead moves inside the period, so the game is played afresh each generation and no commitment across periods is ever assumed or needed.

A.8 Properties of the distorted measure

This appendix defines the risk-sensitive distorted measure Q_t^* of Section 3 and records two of its properties. The measure reweights the worst-case transition law $P_{\rho_t^*, z_t^*}$ with Radon–Nikodym derivative

$$\frac{dQ_t^*}{dP_{\rho_t^*, z_t^*}} = \frac{\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\}}{\mathbb{E}_{\rho_t^*, z_t^*}[\exp\{-\gamma\Phi(n_t^*)V(x_{t+1}, s_{t+1})\} \mid x_t, s_t]}.$$

First, the likelihood reweighting is monotone in continuation value. For two successor states ω_1 and ω_2 ,

$$\frac{(dQ_t^*/dP_{\rho_t^*, z_t^*})(\omega_1)}{(dQ_t^*/dP_{\rho_t^*, z_t^*})(\omega_2)} = \exp\{-\gamma\Phi(n_t^*)[V(\omega_1) - V(\omega_2)]\},$$

so if $V(\omega_1) < V(\omega_2)$, then ω_1 receives relatively more weight under Q_t^* than under $P_{\rho_t^*, z_t^*}$. The distorted measure shifts probability mass toward low-value future states, more aggressively the larger is γ , and collapses back to $P_{\rho_t^*, z_t^*}$ as $\gamma \rightarrow 0$.

Second, the CARA certainty equivalent admits the entropy representation familiar from risk-sensitive and robust control,

$$-\frac{1}{\gamma} \log \mathbb{E}_{P_{\rho_t^*, z_t^*}}[e^{-\gamma Y_{t+1}} \mid x_t, s_t] = \min_{Q \ll P_{\rho_t^*, z_t^*}} \left\{ \mathbb{E}_Q[Y_{t+1} \mid x_t, s_t] + \frac{1}{\gamma} D(Q \| P_{\rho_t^*, z_t^*}) \right\},$$

where $Y_{t+1} = \Phi(n_t^*)V(x_{t+1}, s_{t+1})$ and D denotes relative entropy. The minimizer of this entropy-penalized problem is exactly Q_t^* . The representation shows in what sense Q_t^* acts like a pessimistic belief. It puts extra weight on bad future states, but the parent is not simply maximizing expected utility under Q_t^* . The distorted measure is generated by the risk-sensitive problem itself, changes with the chosen policy, and is disciplined by the entropy cost that limits how far it can move from the baseline worst-case law.

A.9 Risk-neutral and ambiguity-neutral limits

The model differs from the standard Barro–Becker problem in two ways. First, the parent is ambiguity averse. She evaluates each plan under the worst admissible signal-informativeness parameter $\rho \in [\underline{\rho}, \bar{\rho}]$. Second, she is risk sensitive. Future continuation values enter through the CARA certainty equivalent rather than through a standard expectation. This appendix separates the two forces explicitly (see Remark 2 in the main text).

Start by shutting down risk sensitivity. By Lemma 1, $\mathcal{C}_{\gamma, \rho, z}(Y) = \mathbb{E}_{\rho, z}[Y] - \frac{\gamma}{2} \text{Var}_{\rho, z}(Y) + o(\gamma)$, so as $\gamma \rightarrow 0$ the certainty equivalent converges to a conditional expectation. Because $\Phi(n_t)$ is

chosen at time t and is nonnegative, it can then be taken outside both the expectation and the minimization, and the Bellman equation becomes

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \Phi(n_t) \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \mathbb{E}_{\rho, z_t} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\}. \quad (\text{A.1})$$

Equation (A.1) is the risk-neutral robust Barro–Becker problem. Fertility still scales the value of descendants through $\Phi(n_t)$, but the continuation value is evaluated under the worst admissible signal-informativeness parameter.

Now shut down ambiguity instead. If the ambiguity set is a singleton, $\underline{\rho} = \bar{\rho} = \rho$, the inner minimization disappears and the problem is ambiguity-neutral but still risk sensitive. The parent uses a single transition law, indexed by ρ , and evaluates future outcomes through the CARA certainty equivalent in (6). Finally, if both features are switched off, that is $\gamma \rightarrow 0$ and $\underline{\rho} = \bar{\rho} = \rho$, the problem collapses to the standard Barro–Becker recursive formulation

$$V(x_t, s_t) = \max_{c_t, n_t, z_t} \left\{ u(c_t) + \beta \Phi(n_t) \mathbb{E} [V(x_{t+1}, s_{t+1}) \mid x_t, s_t] \right\}. \quad (\text{A.2})$$

Thus the standard model is recovered only when the parent uses a single probability law and evaluates continuation value by an ordinary expectation.

B Robustness of the Inertia Result

This appendix states and proves the robustness results summarized in Section 4.7.

B.1 Smooth-ambiguity sketch

Let $I = [\rho, \bar{\rho}]$ and let π be a probability measure on I . Under exponential smooth ambiguity, the parent evaluates a plan $\theta = (n, z)$ by

$$V_\eta(\theta, s) = -\frac{1}{\eta} \log \mathbb{E}_\pi \left[e^{-\eta K(\theta, \rho, s)} \right], \quad \eta > 0,$$

where K is the LQ certainty equivalent (18). The second-order belief π is held fixed as s varies.

We show that the maxmin inertia result is recovered as ambiguity aversion becomes large. Let S be a compact subset of the maxmin interior-belief region and let N be a compact neighborhood of the signal-free optimizer θ_0 . Suppose that, on $N \times S$,

$$A(\theta) \equiv \Phi(n)\sigma(z)$$

is bounded away from zero and

$$\rho^w(\theta, s) = -\frac{s}{\gamma A(\theta)}$$

stays a positive distance from the endpoints of I . Assume that π has a positive, sufficiently smooth density p near these minimizers and that the signal-free outer Hessian at θ_0 is nonsingular. Then there is, for all sufficiently large η , a local optimizer branch $\theta_\eta^*(s)$ near θ_0 satisfying

$$\sup_{s \in S} \left\| \frac{\partial \theta_\eta^*(s)}{\partial s} \right\| = O(\eta^{-1}).$$

Thus the exactly flat maxmin policy is the large- η limit of an attenuated smooth-ambiguity response.

To prove the claim, first define the ambiguity-adjusted decision weight

$$\frac{d\pi_{\eta, \theta, s}}{d\pi}(\rho) = \frac{e^{-\eta K(\theta, \rho, s)}}{\mathbb{E}_\pi[e^{-\eta K(\theta, \rho, s)}]}.$$

Since $K_s = A(\theta)\rho$, differentiation under the integral gives

$$\frac{\partial}{\partial s} \mathbb{E}_{\pi_{\eta, \theta, s}}[\rho] = -\eta A(\theta) \text{Var}_{\pi_{\eta, \theta, s}}(\rho) \leq 0.$$

Hence a lower signal shifts weight toward larger values of ρ , the same direction as the maxmin selector $\rho^w(\theta, s)$.

On the interior-belief region, completing the square yields

$$K(\theta, \rho, s) = M(\theta, s) + \frac{\gamma A(\theta)^2}{2} (\rho - \rho^w(\theta, s))^2,$$

where

$$M(\theta, s) = \min_{\rho \in I} K(\theta, \rho, s) = \Phi(n)\mu(z) - \frac{\gamma}{2} A(\theta)^2 - \frac{s^2}{2\gamma}.$$

Laplace's method, uniformly on $N \times S$ and through the derivatives used below, gives

$$V_\eta(\theta, s) = M(\theta, s) + \frac{\log \eta}{2\eta} + \frac{1}{2\eta} \log \left(\frac{\gamma A(\theta)^2}{2\pi} \right) - \frac{1}{\eta} \log p(\rho^w(\theta, s)) + o(\eta^{-1}).$$

This expansion first shows that V_η converges uniformly to the maxmin value M . More importantly, $M_{\theta s} \equiv 0$ and the other s -dependent term is multiplied by η^{-1} , so

$$V_{\eta, \theta s} = -\frac{1}{\eta} \partial_{\theta s} \log p(\rho^w(\theta, s)) + o(\eta^{-1}) = O(\eta^{-1}).$$

For the outer objective

$$F_\eta(\theta, s) = u(m_t - n(q_t + z)) + \beta V_\eta(\theta, s),$$

the implicit function theorem supplies the nearby optimizer branch and a uniformly bounded inverse Hessian. Differentiating its first-order condition therefore gives

$$\frac{\partial \theta_\eta^*}{\partial s} = -F_{\eta, \theta\theta}^{-1} \beta V_{\eta, \theta s} = O(\eta^{-1}),$$

which proves the stated attenuation result. This is a local result on compact subsets of the interior-belief region; it does not impose a global shape on the finite- η policy.

B.2 Discrete fertility

Because fertility is an integer, a small change in the signal need not change the chosen family size. Ambiguity gives a stronger result. Compare two adjacent positive family sizes, n and $n+1$. When both plans have an interior worst-case belief, the signal changes their values by the same amount. Their value difference is therefore constant, so the parent cannot switch between them in that region. This argument does not apply to the childless plan: choosing $n=0$ gives no continuation value and hence no common signal penalty.

Let $y \geq 0$. The effective cost per child is $q_e > 0$: it equals q without a subsidy and $q - \underline{\alpha}\tau$ with the policy. The feasible family sizes are

$$\mathcal{N}_y = \left\{ 0, 1, \dots, \left\lfloor \frac{y}{q_e} \right\rfloor \right\}.$$

For $n \in \mathcal{N}_y$, write total robust value as

$$W_n(s) = y - q_e n + \beta \min_{\rho \in [\underline{\rho}, \bar{\rho}]} \left\{ n(\mu + \sigma \rho s) - \frac{\gamma}{2} n^2 \sigma^2 (1 - \rho^2) \right\},$$

with $W_0(s) = y$.

Proposition B.2.1 (Discrete fertility). *Assume $0 \leq \underline{\rho} < \bar{\rho} < 1$ and $\beta, \gamma, \sigma, q_e > 0$. Then the following statements hold.*

(i) *Fix a signal s and allow n to vary continuously over $[0, y/q_e]$. The resulting value function is strictly concave. Thus there is one optimal integer, except possibly when two adjacent integers are tied.*

(ii) *Let $n \geq 1$ and suppose $n, n+1 \in \mathcal{N}_y$. Define*

$$C_n = [-\gamma \sigma \bar{\rho} n, -\gamma \sigma \underline{\rho} (n+1)].$$

The interval C_n has positive length if and only if

$$\frac{\underline{\rho}}{\bar{\rho}} < \frac{n}{n+1}.$$

On all of C_n ,

$$W_{n+1}(s) - W_n(s) = c_n \equiv \beta \left[\mu - \frac{\gamma}{2} \sigma^2 (2n+1) \right] - q_e. \quad (\text{B.1})$$

Inside C_n , both plans have an interior worst-case belief. At an endpoint, one belief reaches a bound, but the value formula remains valid by continuity. If $c_n \neq 0$, the same plan is preferred at every signal in C_n . If $c_n = 0$, the two plans are tied throughout C_n .

(iii) The difference $\Delta_n(s) = W_{n+1}(s) - W_n(s)$ never decreases as s rises. If $\underline{\rho} > 0$ and C_n has positive length, it rises strictly outside C_n . Under these conditions, the two plans have a unique switching point when $c_n \neq 0$. It lies to the left of C_n if $c_n > 0$ and to the right if $c_n < 0$. If $c_n = 0$, the plans are tied on all of C_n .

Proof. For $n \geq 1$, completing the square in ρ gives

$$\begin{aligned} n(\mu + \sigma \rho s) - \frac{\gamma}{2} n^2 \sigma^2 (1 - \rho^2) \\ = n\mu - \frac{\gamma}{2} n^2 \sigma^2 - \frac{s^2}{2\gamma} + \frac{\gamma}{2} n^2 \sigma^2 \left(\rho + \frac{s}{\gamma n \sigma} \right)^2. \end{aligned} \quad (\text{B.2})$$

Thus the unique worst-case belief is

$$\rho_n^w(s) = \text{Proj}_{[\underline{\rho}, \bar{\rho}]} \left(-\frac{s}{\gamma n \sigma} \right),$$

and the minimized continuation value is

$$\underline{K}_n(s) = \begin{cases} n\mu + n\sigma \bar{\rho} s - \frac{\gamma}{2} n^2 \sigma^2 (1 - \bar{\rho}^2), & s \leq -\gamma n \sigma \bar{\rho}, \\ n\mu - \frac{\gamma}{2} n^2 \sigma^2 - \frac{s^2}{2\gamma}, & -\gamma n \sigma \bar{\rho} \leq s \leq -\gamma n \sigma \underline{\rho}, \\ n\mu + n\sigma \underline{\rho} s - \frac{\gamma}{2} n^2 \sigma^2 (1 - \underline{\rho}^2), & s \geq -\gamma n \sigma \underline{\rho}. \end{cases} \quad (\text{B.3})$$

The three pieces meet with the same first derivative. For fixed s , the second derivative of the continuous value function is one of

$$-\beta \gamma \sigma^2 (1 - \bar{\rho}^2), \quad -\beta \gamma \sigma^2, \quad -\beta \gamma \sigma^2 (1 - \underline{\rho}^2).$$

Each expression is negative, so the continuous value function is strictly concave. It can therefore have at most two optimal integers, and two can be optimal only if they are adjacent. This proves part (i).

For plan n , the unconstrained minimizer belongs to $[\underline{\rho}, \bar{\rho}]$ exactly when

$$-\gamma \sigma \bar{\rho} n \leq s \leq -\gamma \sigma \underline{\rho} n.$$

The same condition for plan $n+1$ gives another interval. Their overlap is C_n , which has positive length exactly when $\bar{\rho} n > \underline{\rho} (n+1)$. On C_n , the middle line of (B.3) applies to both plans, with the endpoint values included by continuity. Both values contain the term $-\beta s^2/(2\gamma)$, so it cancels when the plans are compared:

$$\begin{aligned} W_{n+1}(s) - W_n(s) &= -q_e + \beta \left[\mu - \frac{\gamma}{2} \sigma^2 ((n+1)^2 - n^2) \right] \\ &= \beta \left[\mu - \frac{\gamma}{2} \sigma^2 (2n+1) \right] - q_e. \end{aligned}$$

This proves part (ii).

For part (iii), the envelope theorem gives

$$\frac{\partial W_n(s)}{\partial s} = \beta n \sigma \rho_n^w(s), \quad n \geq 1.$$

Let $x = -s/(\gamma \sigma)$. Rescaling the projection gives

$$n \rho_n^w(s) = \text{Proj}_{[n\underline{\rho}, n\bar{\rho}]}(x).$$

It follows that

$$\begin{aligned} \Delta'_n(s) = \beta\sigma \Big[& \text{Proj}_{[(n+1)\underline{\rho}, (n+1)\bar{\rho}]}(x) \\ & - \text{Proj}_{[n\underline{\rho}, n\bar{\rho}]}(x) \Big] \geq 0. \end{aligned} \quad (\text{B.4})$$

The first projection interval has larger lower and upper endpoints than the second, so the difference is nonnegative. If $\underline{\rho} > 0$ and the intervals overlap, the difference is zero exactly when $x \in [(n+1)\underline{\rho}, n\bar{\rho}]$, or equivalently when $s \in C_n$. Also,

$$\lim_{s \rightarrow -\infty} \Delta_n(s) = -\infty, \quad \lim_{s \rightarrow +\infty} \Delta_n(s) = +\infty$$

when $\underline{\rho} > 0$. Thus $\Delta_n(s)$ moves continuously from negative to positive values, rises strictly outside C_n , and equals c_n on C_n . This proves the threshold statements. $\square$

The condition $n \geq 1$ matters. Because $W_0(s) = y$, the childless plan has no common signal penalty. When the one-child plan has an interior worst-case belief,

$$W_1(s) - W_0(s) = \beta \left[\mu - \frac{\gamma}{2} \sigma^2 \right] - q_e - \frac{\beta s^2}{2\gamma}. \quad (\text{B.5})$$

This difference depends on s and may equal zero inside the one-child interior region. Thus Proposition B.2.1 rules out switching inside a common-interior region only when both family sizes are positive.

The result compares one pair of plans; it does not by itself identify the global optimum. Another family size may become relevant inside C_n . But when n and $n+1$ are the two relevant choices, (B.1) fixes their ranking on C_n , even though the value of each plan changes with the signal. Welfare can therefore move while the chosen family size does not.

For comparison, with known $\rho \in (0, 1)$ the adjacent value difference is

$$-q_e + \beta \left[\mu + \sigma \rho s - \frac{\gamma}{2} \sigma^2 (1 - \rho^2) (2n + 1) \right],$$

so its unique threshold is

$$s_n(\rho) = \frac{q_e/\beta - \mu + \frac{\gamma}{2} \sigma^2 (1 - \rho^2) (2n + 1)}{\sigma \rho},$$

and the width of an unconstrained interior band is

$$s_n(\rho) - s_{n-1}(\rho) = \gamma \sigma \frac{1 - \rho^2}{\rho}.$$

This width falls as ρ rises. In the ambiguity model, if both plans choose the upper bound $\bar{\rho}$, their switching point is the same as in the known- $\bar{\rho}$ model. The same is true at the lower bound $\underline{\rho}$. Ambiguity therefore does not make every integer-choice band wider than every known- ρ benchmark. In a transition region, one plan may choose a boundary belief while the other still chooses an interior belief, so no single known- ρ formula describes their comparison.

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